



Enhanced tumour penetration and prolonged circulation in blood of polyzwitterion–drug conjugates with cell-membrane affinity

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Effective anticancer nanomedicines need to exhibit prolonged circulation in blood, to extravasate and accumulate in tumours, and to be taken up by tumour cells. These contrasting criteria for persistent circulation and cell-membrane affinity have often led to complex nanoparticle designs with hampered clinical translatability. Here, we show that conjugates of small-molecule anticancer drugs with the polyzwitterion poly(2-(*N*-oxide-*N,N*-diethylamino)ethyl methacrylate) have long blood-circulation half-lives and bind reversibly to cell membranes, owing to the negligible interaction of the polyzwitterion with proteins and its weak interaction with phospholipids. Adsorption of the polyzwitterion–drug conjugates to tumour endothelial cells and then to cancer cells favoured their transcytosis-mediated extravasation into tumour interstitium and infiltration into tumours, and led to the eradication of large tumours and patient-derived tumour xenografts in mice. The simplicity and potency of the polyzwitterion–drug conjugates should facilitate the design of translational anticancer nanomedicines.

Nanomedicine has been extensively explored as a candidate strategy that could revolutionize cancer treatments^{1–3}. Several nanomedicines such as albumin-bound paclitaxel⁴ and liposome-encapsulated irinotecan^{5,6} can mitigate the drugs' adverse effects and enhance therapeutic efficacy through a complicated delivery process^{7,8}. However, the therapeutic efficacies of anticancer nanomedicines remain far below expectations¹.

Cancer drug delivery has to overcome a series of complex physiological and pathological barriers^{7–12}. In general, non-stickiness towards cells and proteins (or 'stealth') is considered essential for nanomedicines to remain in the blood for a sufficient duration and to infiltrate into tumour tissue¹³, whereas tumour extravasation from blood capillaries and cell internalization steps require the nanomedicines to interact with the cell membrane¹⁰. Thus, the nanomedicine surface is generally engineered to undergo a stealthy-to-sticky transition¹⁰. Nanomedicines are made stealthy in the blood circulation by masking their surface with antifouling polymers¹⁴ (for example, polyethylene glycol (PEG)¹⁵ and polyzwitterions¹⁶) or biomimetic leukocyte membranes^{17–19}, and are then functionalized with targeting or binding ligands to restore their cell-binding ability^{6,10,20,21}. Thus, nanomedicines are integrated with different functional components or moieties for these functions^{22–26}. However, there are difficulties in characterizing each component's *in vivo* behaviours and establishing production processes compliant with good manufacturing practices for such complicated nanomedicines, and thus few

get to the clinical translation stage^{26–29}. Therefore, a distinct design strategy able to simplify the structures but retain these functions is key to developing translational nanomedicines.

We recently demonstrated that rather than making the nanomedicine surface stealthy to promote tumour penetration¹³, nanocarriers binding to cells could exploit adsorption-mediated transcytosis (AMT) for effective tumour extravasation and penetration³⁰. Thus, we further hypothesized that a nanomedicine that was not sticky towards proteins but would bind to cells would be capable of efficient cancer drug delivery. Non-stickiness towards proteins generally prolongs a nanomedicine's lifetime in circulation in the blood³¹; binding to and thus 'hitchhiking' on red blood cells (RBCs) also enables nanomedicines to circulate for longer³²; binding to tumour capillary endothelial cells (tECs) and, subsequently, tumour cells facilitates extravasation and tumour penetration by transcytosis^{30,33}. Key aspects of such a design are non-stickiness towards proteins and a proper cell-binding affinity, because adsorption of bloodborne proteins quickly induces opsonization and rapid clearance by the immune system^{34,35}, and strong binding to cells causes blood clots and aggregation of RBCs³⁶ and makes the nanomedicine unable to translocate to tECs for extravasation³².

Here we show that poly(2-(*N*-oxide-*N,N*-diethylamino)ethyl methacrylate) (OPDEA), a phospholipid-binding zwitterion that is not sticky towards proteins, is capable of efficient cancer drug delivery with high therapeutic efficacy (Fig. 1). Poly(*N*-oxide) is

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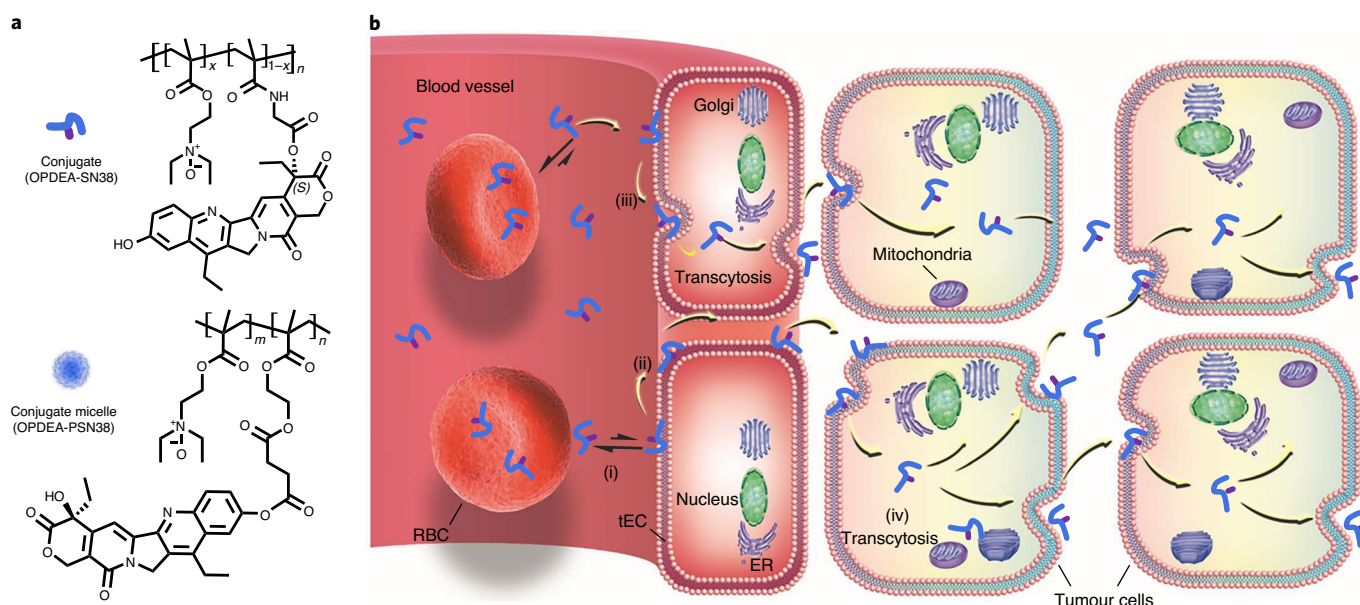


Fig. 1 | OPDEA-based conjugate or block copolymer with the anticancer drug SN38 and the drug delivery process. **a**, Molecular structures of the OPDEA conjugate (OPDEA-SN38) and conjugate block copolymer (OPDEA-PSN38); the block copolymer forms conjugate micelles in aqueous solution. **b**, After intravenous injection, the OPDEA conjugate or micelles circulate in the plasma or by reversibly attaching to RBCs, and may also adsorb on the tECs (i); adsorption on the tECs facilitates the extravasation of the conjugate or micelles via the EPR effect (ii) and may also trigger AMT through the tECs into the interstitium (iii). In the tumour interstitium, adsorption on the cancer cell initiates rapid internalization of the nanomedicine, some of which is exocytosed and internalized by the surrounding cells and passed on to cells in deeper layers, leading to infiltration into the avascular regions of the tumour (iv). ER, endoplasmic reticulum.

not sticky towards proteins³⁷, but we have found that the *N*-oxide moiety interacts with the hydrophilic heads of phosphatidylcholine (PC) and phosphatidylethanolamine (PE), thus reversibly binding to cell membranes. After intravenous injection, OPDEA circulates in the blood, in dynamic equilibrium between the plasma and the surface of RBCs. OPDEA can bind tECs (Fig. 1b(i)), facilitating its extravasation into tumour interstitium via the enhanced permeability and retention³⁸ (EPR) effect (Fig. 1b(ii)) or by triggering fast AMT (Fig. 1b(iii)) for active extravasation. In the tumour interstitium, its binding on cancer cells triggers efficient transcytosis-based intercellular delivery from perivascular to avascular and hypoxic areas (Fig. 1b(iv)). Together, these effects enable OPDEA to deliver drugs efficiently to all tumour cells, eradicating exponentially growing large (approximately 500 mm³) tumours as well as clinically relevant xenograft tumours derived from human patients.

Results

Synthesis of OPDEA, OPDEA-SN38 and OPDEA-PSN38.

OPDEA and the related conjugates were synthesized and characterized as shown in Supplementary Scheme 1 and Supplementary Figs. 1–13. Living atom-transfer radical polymerization (ATRP) of 2-(*N,N*-diethylamino)ethyl methacrylate (DEA) was used to prepare the DEA polymer, PDEA. The molecular mass of PDEA was deliberately controlled to be around 20 kDa, lower than the molecular mass cut-off of renal clearance. The PDEA was then oxidized to OPDEA with hydrogen peroxide at room temperature.

OPDEA is soluble in water, methanol, and *N,N*-dimethylformamide (DMF) but insoluble in low-polarity organic solvents such as acetone, chloroform and tetrahydrofuran (THF). To conjugate the potent anticancer drug 7-ethyl-10-hydroxycamptothecin (SN38) to OPDEA, carboxylic acid groups were introduced to the OPDEA chains by copolymerization of DEA with *tert*-butyl methacrylate followed by oxidation and deprotection (Supplementary Figs. 3, 4 and 6–9). Finally, 3.7 wt% of SN38 was conjugated via its stable 20-OH

ester to OPDEA to produce the drug conjugate OPDEA-SN38 (Supplementary Fig. 9). OPDEA was also labelled with 1.0 wt% of cyanine5.5 NHS ester (Cy5.5) to obtain fluorescently labelled Cy5.5-OPDEA (Supplementary Scheme 1). Linear methoxyl-PEG (20 kDa) conjugated with SN38 (PEG-SN38) was synthesized as the control (Supplementary Fig. 10).

We also synthesized the amphiphilic block copolymer of OPDEA with an SN38-containing polymer chain as the hydrophobic block to fabricate drug-conjugate micelles (Supplementary Scheme 1). SN38 was conjugated through a succinic acid linker to 2-hydroxyethyl methacrylate to obtain the SN38 monomer (SN38-MA)³⁹. We used the phenol (10-OH) ester of SN38 as the conjugation site because the phenol ester readily hydrolyzes and releases free SN38, particularly in the presence of high levels of glutathione in tumours^{39,40}. We performed successive block copolymerization of DEA with SN38-MA using a typical ATRP procedure³⁹. The conversion of SN38-MA in the first block polymerization was controlled to over 95%. The resulting poly(SN38-MA) (PSN38) had a polydispersity index (PDI) of 1.4 because of the high monomer conversion (Supplementary Fig. 11). DEA was then added for the subsequent polymerization to produce PDEA-block-PSN38 (PDEA-PSN38), and both block lengths were controlled to be 20 kDa (Supplementary Figs. 11–13). Subsequent oxidation of the PDEA block in DMF using *meta*-chloroperoxybenzoic acid as an oxidant for 0.5 h produced OPDEA-PSN38 without hydrolysis of the phenol esters or SN38 oxidation. The block copolymer PEG-PSN38 as control was also synthesized via ATRP using methoxyl-PEG as the macroinitiator³⁹. The block copolymers were labelled by the thiolysis reaction of the SN38 phenol ester with thiol-functionalized Cy5.5 or rhodamine B (RhoB)⁴⁰.

OPDEA and OPDEA-SN38 (3.7 wt% SN38) were water-soluble and nearly neutral at physiological pH (Supplementary Fig. 14). The block copolymer OPDEA-PSN38 formed well-defined micelles of about 51 nm in diameter and about −10.5 mV in ζ -potential

(Supplementary Fig. 15). PEG-PSN38 micelles had a mean diameter of 35 nm with a ζ -potential of -11.1 mV.

OPDEA undergoes fast cellular uptake via macropinocytosis. Zwitterionic polymers are neutral, ultra-hydrophilic and generally have ultra-low fouling—that is, low interactions with proteins and cells—thus having slow cellular internalization and low cytotoxicity⁴¹. OPDEA did not affect the cell morphology and had low cytotoxicity *in vitro* and *in vivo*, even at concentrations as high as $100\text{ }\mu\text{g ml}^{-1}$ (Supplementary Fig. 16). Cancer cells rapidly took up $\text{Cy}5.5$ -OPDEA, with more than 100-fold of the uptake of $\text{Cy}5.5$ -PEG at 15 or 60 min (Fig. 2a and Supplementary Fig. 17a,b), as shown by time-lapse imaging (Supplementary Videos 1 and 2). Flow cytometry analysis (Fig. 2b) showed that after 1 h incubation, 25.6% of cells treated with $\text{Cy}5.5$ -OPDEA, but only 6.3% of cells treated with $\text{Cy}5.5$ -PEG, were positive for Cy5.5. These results indicated that despite being zwitterionic, OPDEA entered cells much more rapidly than PEG with the same molecular mass (20 kDa).

We further analysed the cellular uptake pathway and subcellular distribution of $\text{Cy}5.5$ -OPDEA. At different incubation times, $\text{Cy}5.5$ -OPDEA was found to be not associated with lysosomes stained with LysoTracker green (Fig. 2a and Supplementary Fig. 17c), whereas $\text{Cy}5.5$ -PEG was mostly confined to lysosomes (Supplementary Fig. 17d). Whether the non-lysosomal localization of $\text{Cy}5.5$ -OPDEA was due to fast lysosomal escape by lysis or the non-lysosomal endocytosis pathway was probed using acridine orange, an acid-sensitive dye that emits red fluorescence in acidic lysosomes but becomes green once released into the cytosol⁴². As a control, chloroquine lysed lysosomes, and thus cells treated with chloroquine showed weak acridine orange fluorescence (Supplementary Fig. 18), whereas the red fluorescence in OPDEA-treated HepG2 human hepatocellular carcinoma cells was as strong as those of the control and PEG-treated cells, indicating that the lysosomes in the OPDEA-treated cells were well preserved. An RBC lysis assay, generally used to evaluate a polymer's ability to rupture lysosomes⁴³, showed that OPDEA did not cause RBC lysis (Supplementary Fig. 19). OPDEA therefore did not rupture lysosomes, and accordingly, $\text{Cy}5.5$ -OPDEA was not internalized into lysosomes. Time-lapse imaging also showed no $\text{Cy}5.5$ -OPDEA signal overlapping with lysosomes following addition of $\text{Cy}5.5$ -OPDEA to the cells (Supplementary Video 1).

The cellular uptake pathway study showed that a low temperature of 4°C completely inhibited the cellular uptake of OPDEA, indicating an energy-dependent endocytosis pathway (Supplementary Fig. 20a). Heparin, which inhibits cell macropinocytosis via heparan sulfate proteoglycan (a macropinocytosis receptor)⁴⁴, prevented OPDEA internalization (Supplementary Fig. 20a). The endocytosis inhibitors filipin (an inhibitor of caveolae-mediated endocytosis) and chlorpromazine (an inhibitor of clathrin-mediated endocytosis) did not affect OPDEA uptake, but wortmannin (an inhibitor of phosphoinositide 3-kinase) and cytochalasin D (an inhibitor of actin polymerization) slowed the internalization (Supplementary Fig. 20b). These results strongly suggested that OPDEA entered tumour cells through the macropinocytosis-related pathway. Tetramethylrhodamine-labelled neutral dextran (TRITC -Dex, 70 kDa), a macropinosome marker, was further used to track the colocalization of $\text{Cy}5.5$ -OPDEA in macropinosomes. Fluorescent signals of $\text{Cy}5.5$ -OPDEA and TRITC -Dex overlapped at observed time points (Fig. 2c). In the linescan analysis, the $\text{Cy}5.5$ -OPDEA (red lines) and TRITC -Dex (blue lines) showed good colocalization (Fig. 2d), confirming the localization of $\text{Cy}5.5$ -OPDEA in macropinosomes. Similarly, OPDEA- $\text{Cy}5.5$ -PSN38 micelles were also internalized quickly via the macropinocytosis pathway and colocalized with macropinosomes (Supplementary Fig. 21).

OPDEA conjugates and micelles undergo slow blood clearance. Figure 3a compares the blood clearance of intravenously injected OPDEA-SN38 conjugate with that of PEG-SN38. The overall

clearance kinetics of OPDEA-SN38 was similar to those of PEG-SN38, but OPDEA-SN38 seemed faster in distribution and slower in later clearance than PEG-SN38, as seen from their distribution half-life time ($t_{\alpha/2}$) and elimination half-life time ($t_{\beta/2}$) (see Supplementary Table 1 for detailed information). The area under the concentration–time curve in the first 12 h ($\text{AUC}_{0-12\text{h}}$) was $37.9\text{ ID}\%\text{ h ml}^{-1}$ for PEG-SN38 and $29.6\text{ ID}\%\text{ h ml}^{-1}$ for OPDEA-SN38. The total SN38-concentration in tumours from the mice injected with OPDEA-SN38 was lower than that from the mice injected with PEG-SN38 6 h after injection, but they were equal at 24 h (Fig. 3b). *In vivo* imaging confirmed tumour accumulation of OPDEA-SN38 and PEG-SN38 (Fig. 3c).

The blood clearance and calculated pharmacokinetic $t_{\alpha/2}$ and $t_{\beta/2}$ values of OPDEA-PSN38 and PEG-PSN38 micelles are shown in Fig. 3d and Supplementary Table 1. OPDEA-PSN38 micelles had a half-life in the blood of 8.2 h, about half that of PEG-PSN38 micelles. After 24 h, 1.8% of OPDEA-PSN38 micelles and 7.0% of PEG-PSN38 micelles remained in the blood. The corresponding $\text{AUC}_{0-12\text{h}}$ values were $300\text{ ID}\%\text{ h ml}^{-1}$ for PEG-PSN38 micelles and $158\text{ ID}\%\text{ h ml}^{-1}$ for OPDEA-PSN38 micelles. Meanwhile, the total SN38 concentrations in the mouse urine at 6 h and 24 h were $0.52 \pm 0.08\%$ and $0.16 \pm 0.01\%$ of the injected dose, respectively (Supplementary Fig. 22a,b), indicating that few micelles were excreted into the urine. *In vivo* imaging (Supplementary Fig. 22c) also showed that the mice injected with PEG- $\text{Cy}5.5$ -PSN38 micelles exhibited more widespread fluorescence signals in the body even after 24 h, suggesting that PEG- $\text{Cy}5.5$ -PSN38 is more persistent in the blood circulation than OPDEA- $\text{Cy}5.5$ -PSN38. The accumulation of OPDEA- $\text{Cy}5.5$ -PSN38 in the tumour was already apparent at 30 min after injection (Supplementary Fig. 22c). Biodistribution analysis 8 h after injection (Fig. 3e) showed that OPDEA-PSN38 and PEG-PSN38 micelles exhibited similar low accumulation in heart and lung, but differed in the mononuclear phagocytic system (liver and spleen): OPDEA- $\text{Cy}5.5$ -PSN38 micelles were more easily sequestered by the liver, accounting for their faster blood clearance, whereas PEG- $\text{Cy}5.5$ -PSN38 micelles were taken up by the spleen. However, there was twice as much accumulation of OPDEA- $\text{Cy}5.5$ -PSN38 micelles in tumours as for PEG- $\text{Cy}5.5$ -PSN38 (Fig. 3e).

The time-dependent distributions of OPDEA- $\text{Cy}5.5$ -PSN38 in organs and tumours in HepG2 tumour-bearing mice were further monitored by *ex vivo* imaging (Supplementary Fig. 23a–c). After intravenous injection, OPDEA- $\text{Cy}5.5$ -PSN38 micelles deposited with highest intensity in the liver, second-highest intensity in the kidneys, some in the spleen, lung and intestine, and little in the heart. The intensity of accumulation in the liver decreased quickly over time, and those in the other organs changed slightly, whereas the intensity in the tumour increased over time. Haematoxylin and eosin (H&E) staining of the organs showed no significant morphological changes. Liver and kidney functions were monitored by serum chemistry, and showed no change compared with the healthy control (Supplementary Fig. 23d,e), indicating that OPDEA-PSN38 at this dose did not cause significant toxicity to these organs.

Enhanced tumour extravasation and infiltration of OPDEA. The tumour penetration of OPDEA was first observed using multicellular tumour spheroids (MTSs). MTSs of approximately 1.0 mm in diameter, derived from HepG2 or human colorectal adenocarcinoma (HT29) cell lines, were cultured with $\text{Cy}5.5$ -OPDEA or $\text{Cy}5.5$ -PEG for timed intervals and then freeze-sectioned into $10\text{ }\mu\text{m}$ slices for imaging. The Cy5.5 fluorescence was weak after 4 h incubation with $\text{Cy}5.5$ -PEG both on the spheroid surface and in the interior (Fig. 4a(i)), and did not significantly increase after 6 h culture (Supplementary Fig. 24). HepG2 MTSs treated with $\text{Cy}5.5$ -OPDEA for 4 h had a bright surface and fluorescent interior, indicating that $\text{Cy}5.5$ -OPDEA was highly adsorbed on the surface and some infiltrated the spheroid; this was further confirmed by linescan analysis as a function of

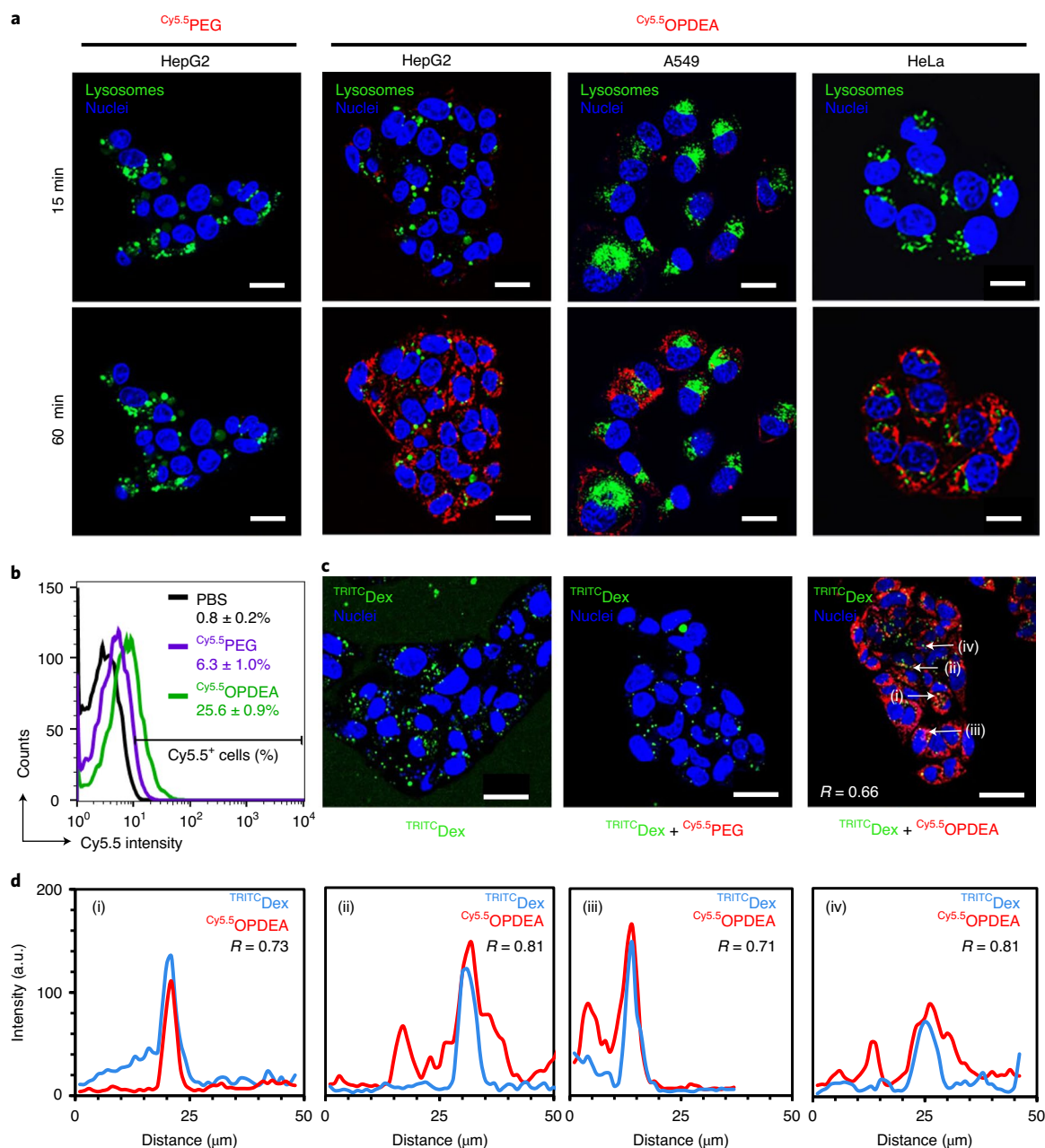


Fig. 2 | In vitro cellular internalization of OPDEA. **a**, Cellular uptake of Cy5.5-OPDEA and Cy5.5-PEG (red) into different cancer cell lines ($n = 3$ biological replicates) and their colocalization with lysosomes (green) observed by confocal laser scanning microscopy (CLSM). Representative images at 15 min and 60 min are shown. Scale bars, 20 μm . Cy5.5 -equivalent dose, $0.5 \mu\text{g ml}^{-1}$. Only one fixed view ($n \geq 8$ cells per sample) could be monitored in time-lapse-acquisition mode; the single stained images of Cy5.5 polymer and their integrated fluorescence intensities are included in Supplementary Fig. 17a,b. Additional images at different time points are included in Supplementary Videos 1 and 2. **b**, Flow cytometry counts showing HepG2 internalization of the labelled polymers. Cy5.5 -equivalent dose, $0.8 \mu\text{g ml}^{-1}$, 1 h culture. A total of 10,000 cells were counted per trial; data are mean $\pm$ s.d. of biological replicates ($n = 3$). **c**, Colocalization of the macropinosome marker dextran (TRITC-Dex , green) and Cy5.5-OPDEA or Cy5.5-PEG (shown in red) in HepG2 cells observed by CLSM ($n = 3$ biological replicates). Cy5.5 -equivalent dose, $0.5 \mu\text{g ml}^{-1}$, 1 h culture. Representative yellow dots indicating colocalization of Cy5.5-OPDEA and TRITC-Dex are indicated by arrows and numbered (i)–(iv). Scale bars, 25 μm . **d**, Linescan analysis by MetaMorph software performed along the randomly selected lines across the numbered dots in **c** to obtain the degree of overlap of Cy5.5-OPDEA (red line) with TRITC-Dex (blue line). Pearson's correlation (R) was used to assess the significance of the correlation between Cy5.5-OPDEA with TRITC-Dex ($n = 4$ biological replicates). a.u., arbitrary units.

distance from the surface to the centre (Fig. 4a(ii) and Supplementary Fig. 24). At 500 μm away from the surface, the fluorescence intensity of HepG2 MTSS treated with Cy5.5-OPDEA was 13–75-fold higher than MTSS treated with Cy5.5-PEG . The overall fluorescence intensity of the Cy5.5-OPDEA -treated HepG2 MTSS increased linearly with culture time, reaching 15 times the intensity of Cy5.5-PEG -treated MTSS

after 4 h (Fig. 4a(iii)). Similar results were observed in HT29 MTSS (Supplementary Fig. 24).

In vivo real-time imaging of Cy5.5-OPDEA and Cy5.5-PEG extravasation from tumour blood vessels into tumour tissue was performed using 4T1 tumours (Supplementary Videos 3–4). After intravenous injection, Cy5.5-PEG was confined in the blood vessels even 120 min

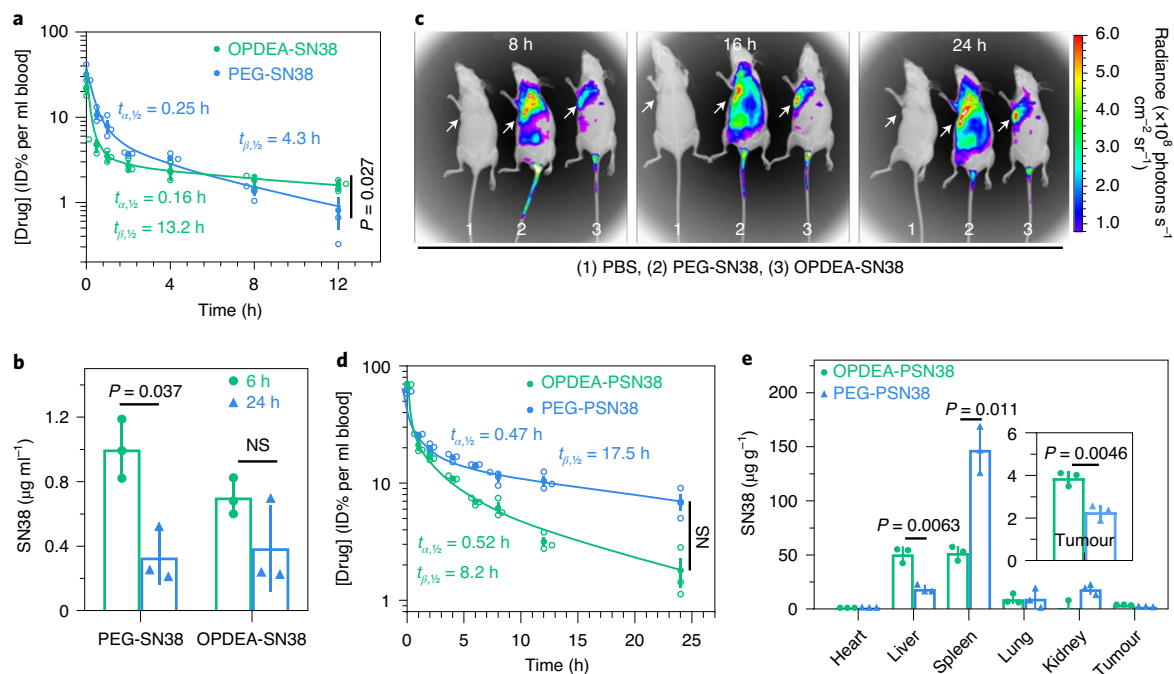


Fig. 3 | In vivo blood clearance of OPDEA-SN38 and PEG-SN38. a,b, Blood clearance kinetics in terms of per cent injected dose (ID%) per ml blood for the conjugates (**a**) and SN38 concentrations in subcutaneous HepG2 tumours at 6 h and 24 h after tail-vein injection (**b**). SN38-equivalent dose, 7.5 mg kg^{-1} . **c**, In vivo fluorescence imaging of nude mice bearing subcutaneous HepG2 tumours at timed intervals after intravenous injection of phosphate-buffered saline (PBS), $\text{Cy}5.5$ PEG or $\text{Cy}5.5$ OPDEA at a $\text{Cy}5.5$ -equivalent dose of 0.5 mg kg^{-1} . Tumour sites are indicated by arrows. **d,e**, Blood clearance (**d**) and total SN38 concentrations in the excised main organs and tumours of mice at 8 h after tail-vein injection of the micelles (**e**). SN38-equivalent dose, 10 mg kg^{-1} . Data are mean $\pm$ s.d. of biological replicates ($n = 3$ mice per treated group); unpaired two-tailed Student's *t*-test. NS, not significant.

after injection, and there was only weak fluorescence in the tumour tissues (Fig. 4b). $\text{Cy}5.5$ OPDEA, however, quickly diffused from the vasculature into the tumour tissue and distributed throughout the whole observed area; the overall fluorescence intensity of a randomly selected area (white dashed frame) increased rapidly in the first 20 min and levelled off gradually, reaching about sixfold of the intensity in the tumour treated with $\text{Cy}5.5$ PEG at 120 min (Fig. 4b), as confirmed by the $\text{Cy}5.5$ OPDEA distribution in images of tumour sections (Supplementary Fig. 25a). Similar results were obtained from subcutaneous and ear-inoculated HepG2 tumour models (Supplementary Fig. 25b–d). Of note, $\text{Cy}5.5$ OPDEA did not extravasate from a normal blood vessel (a subcutaneous venous vessel in the abdomen) (Supplementary Fig. 25e).

We further performed ex vivo analysis of the tumour infiltration of OPDEA by observing its ability to reach avascular hypoxic areas, which are typically more than $100 \mu\text{m}$ away from the vascular network. Nude mice bearing HepG2 tumours of about 10 mm^3 were treated with $\text{Cy}5.5$ OPDEA or $\text{Cy}5.5$ PEG through tail-vein injection. Six hours later, the tumours were dissected and freeze-sectioned into $10\text{-}\mu\text{m}$ -thick sections. Three adjacent tumour sections were separately stained for analysis. In one, $\text{Cy}5.5$ fluorescence was imaged to observe the distribution of the polymer (Fig. 5a(i)). The second was stained with anti-CD31 to label the blood vessels to determine the normoxic regions rich in blood vessels, necrotic regions and avascular hypoxic regions⁴⁵ (Fig. 5a(ii)). The third was stained with H&E for morphological analysis to identify necrotic regions (Fig. 5a(iii)) from the characteristic highly-condensed cell nuclei. In the image taken from the $\text{Cy}5.5$ fluorescence channel, the vasculature-rich normoxic regions, as expected, had strong $\text{Cy}5.5$ fluorescence from the $\text{Cy}5.5$ -labelled polymers, whereas the necrotic regions showed strong autofluorescence. These strong fluorescence signals made it difficult to observe the weak fluorescence in the hypoxic

regions (Supplementary Fig. 26). Manual exclusion of $\text{Cy}5.5$ fluorescence from the normoxic region and autofluorescence from the necrotic region^{46,47} highlighted the $\text{Cy}5.5$ fluorescence in the avascular hypoxic regions. Figure 5a(i–iv) shows that the hypoxic regions in the $\text{Cy}5.5$ OPDEA-treated tumour exhibited much stronger fluorescence signals than those in the $\text{Cy}5.5$ PEG-treated tumour. Linescan analysis of the fluorescence along a randomly selected direction (shown by the arrows in Fig. 5a(i)) indicated that in the $\text{Cy}5.5$ PEG-treated tumour, the fluorescence intensity decreased sharply, showing almost no fluorescence in the hypoxic region; by contrast, the hypoxic region of the $\text{Cy}5.5$ OPDEA-treated tumour exhibited much stronger fluorescence, although this varied, even $300 \mu\text{m}$ away from the normoxic region (Fig. 5a(v)). These results indicated that $\text{Cy}5.5$ OPDEA could penetrate the tumour and reach remote areas.

The distribution of OPDEA- $\text{Cy}5.5$ PSN38 micelles inside the tumour was also analysed by ultrasound and photoacoustic imaging (Fig. 5b). Contrast-enhanced ultrasound was used to create maximum-intensity projections to indicate tumour vascularity—the dark grey area indicates hypovascular tumour (Fig. 5b(i)). The hypoxic area in a tumour can be outlined by tracking oxygenated haemoglobin⁴⁸. A tumour of about 10 mm^3 was used because a tumour of this size has just developed a hypoxic centre, as evidenced by the absence of haemoglobin in the centre of the tumour, and was more easily observed than a larger tumour (Fig. 5b(ii)). Intense fluorescence from OPDEA- $\text{Cy}5.5$ PSN38 was observed in the tumour centre (Fig. 5b(iii)); OPDEA- $\text{Cy}5.5$ PSN38 could penetrate deeply into the tumour's avascular area (Fig. 5b(iv)). This was further verified by in vivo hypoxia staining using intravenous injection of HypoxiSense 680, which binds carbonic anhydrase-9, an enzyme overexpressed by tumour cells in hypoxic regions⁴⁹. OPDEA-PSN38 micelles were labelled with RhoB (OPDEA-RhoBPSN38). Six hours

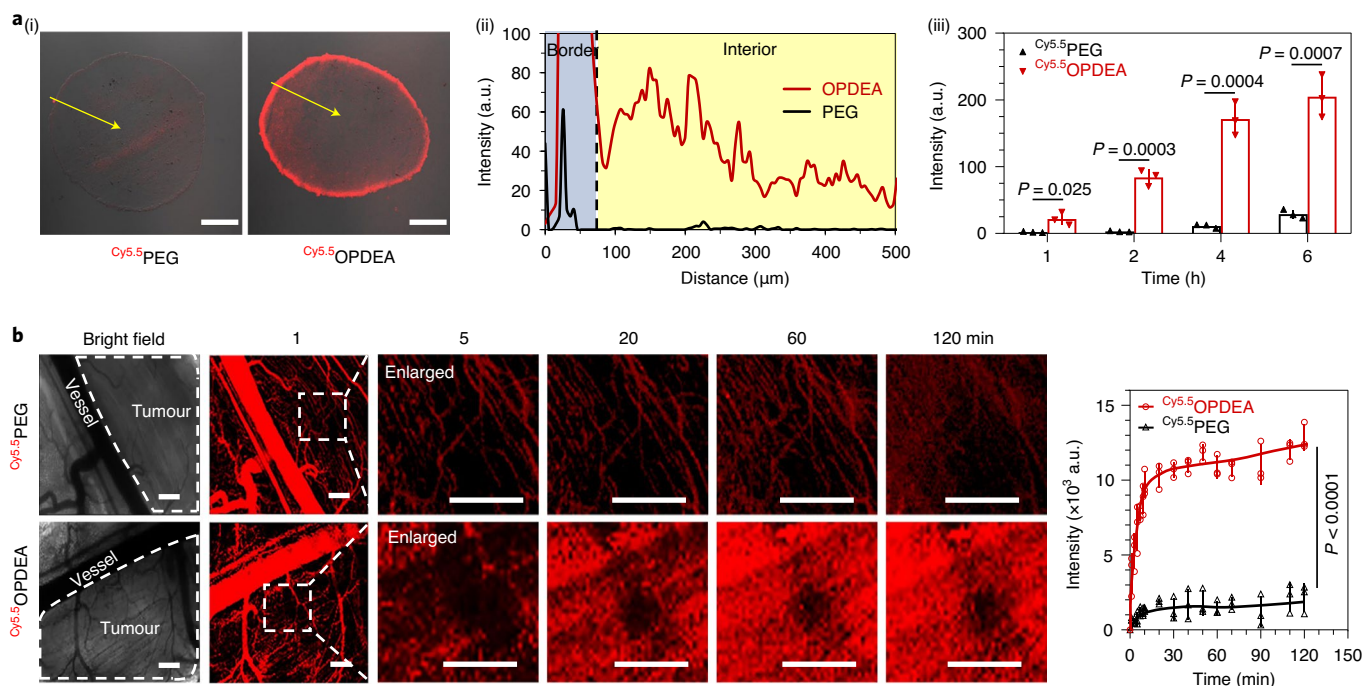


Fig. 4 | In vitro tumour penetration and in vivo tumour extravasation of OPDEA. **a**, Penetration of Cy5.5-OPDEA and Cy5.5-PEG (shown in red) in HepG2 MTSs. MTSs were treated with the polymer (Cy5.5 -equivalent dose, $1.0 \mu\text{g ml}^{-1}$) and imaged by CLSM (see also Supplementary Fig. 24). (i) Representative images of MTS slices after 4 h incubation (3 images obtained). Scale bars, $200 \mu\text{m}$. (ii) Cy5.5 fluorescence intensity from the MTS surface to the centre plotted along randomly selected lines (yellow arrow). (iii) Cy5.5 integrated fluorescence intensity of MTSs at different times. Data are mean $\pm$ s.d. of biological replicates ($n=3$); one-way analysis of variance (ANOVA) with Tukey's correction and 95% confidence intervals. **b**, In vivo fluorescence imaging of extravasation and distribution of Cy5.5-OPDEA or Cy5.5-PEG (red) in a subcutaneous 4T1 tumour after tail-vein injection of the polymer in mice at a Cy5.5 -equivalent dose of 0.5 mg kg^{-1} . Representative images and magnified images of a randomly selected site (dotted outline) are shown (left), and the mean total fluorescence intensity of the selected area was plotted (right). Scale bars, $500 \mu\text{m}$. Only one fixed view could be monitored in time-lapse-acquisition mode; results from other tumour models ($n=3$ biological replicates) are shown in Supplementary Fig. 25b–d; additional images from different time points are included in Supplementary Videos 3 and 4. Data are mean $\pm$ s.d. of biological replicates ($n=3$); unpaired two-tailed Student's *t*-test.

after injection of OPDEA-RhoB-PSN38 , the tumour was dissected, sectioned and observed by CLSM (Fig. 5c). The necrotic area was identified by the characteristic intercellular-space dilatation and cellular atrophy (Fig. 5c(i)). The HypoxiSense 680-stained (green) hypoxic area surrounded the necrotic region (Fig. 5c(ii)), whereas OPDEA-RhoB-PSN38 (red) was distributed throughout the tumour (Fig. 5c(iii)), partially overlapping with HypoxiSense 680 (yellow dots) (Fig. 5c(iv)), confirming that OPDEA-RhoB-PSN38 was able to reach the hypoxic region. By contrast, PEG-Cy5.5-PSN38 showed less extravasation from blood vessels, with almost no signal in the hypoxic regions of the tumour (Supplementary Fig. 27).

OPDEA-PSN38 displays enhanced anticancer activity. The internalized micelles delivered SN38 to the nuclei to exert their action (Supplementary Fig. 28a). OPDEA-SN38 conjugate and OPDEA-PSN38 micelles did not differ in cytotoxicity from their PEG-cojugated counterparts in 2D cell culture (Supplementary Fig. 28b), as indicated by a methylthiazolyldiphenyl-tetrazolium bromide (MTT) assay. However, OPDEA-SN38 conjugate and micelles showed higher cytotoxicity towards 3D MTSs (Fig. 6a).

MTSs treated with OPDEA-PSN38 micelles or PEG-PSN38 showed clear morphological differences (Fig. 6b and Supplementary Fig. 29). Control MTSs treated with PBS were large (diameter $\approx 1,000 \mu\text{m}$) with deeply stained nuclei. PEG-PSN38 -treated MTSs had a loose external layer and a small but hard interior (diameter $\approx 800 \mu\text{m}$) that could be sectioned. By contrast, the MTSs treated with OPDEA-PSN38 micelles were smaller (diameter $\approx 600 \mu\text{m}$)

and too fragile to section, falling apart during transfer; the remaining tumour cells had broken cytoplasm with few lightly stained nuclei (Supplementary Fig. 29). This indicated that OPDEA-PSN38 micelles could induce death of the tumour cells throughout the MTS, whereas PEG-PSN38 killed only the cells in the outer layer, consistent with the results of the penetration experiment (Fig. 5).

We performed preliminary experiments to evaluate the in vivo anticancer activities of the conjugates and micelles. OPDEA alone had no therapeutic efficacy in the HT29 and HepG2 xenograft models (Supplementary Fig. 30a,b). OPDEA-SN38 showed improved efficacy over PEG-SN38 but not as high as expected (Supplementary Fig. 30c), given its increased tumour penetration. This is probably caused by inefficient drug release, owing to the stability of the conjugation via the 20-OH ester bond³⁷. By contrast, the 10-OH (that is, phenol) ester of SN38 is easy to hydrolyze, particularly in tumour cells rich in glutathione³⁹, and the OPDEA-PSN38 micelles accordingly showed much higher antitumour efficacy. We therefore focused on micelles in the subsequent in vivo anticancer experiments.

The micelles' in vivo anticancer activity was first evaluated using small tumours of about 70 mm^3 . At an SN38-equivalent dose of 10 mg kg^{-1} , PEG-PSN38 micelles slowed tumour growth compared with the PBS control, and OPDEA-PSN38 reduced the tumour size after the third intravenous injection (Fig. 6c). After the fifth treatment with OPDEA-PSN38 , five of the six mice showed no sign of tumours (Fig. 6c, inset), and one tumour-bearing mouse had a small tumour of less than 4 mm^3 . Sixty days after the first treatment, the

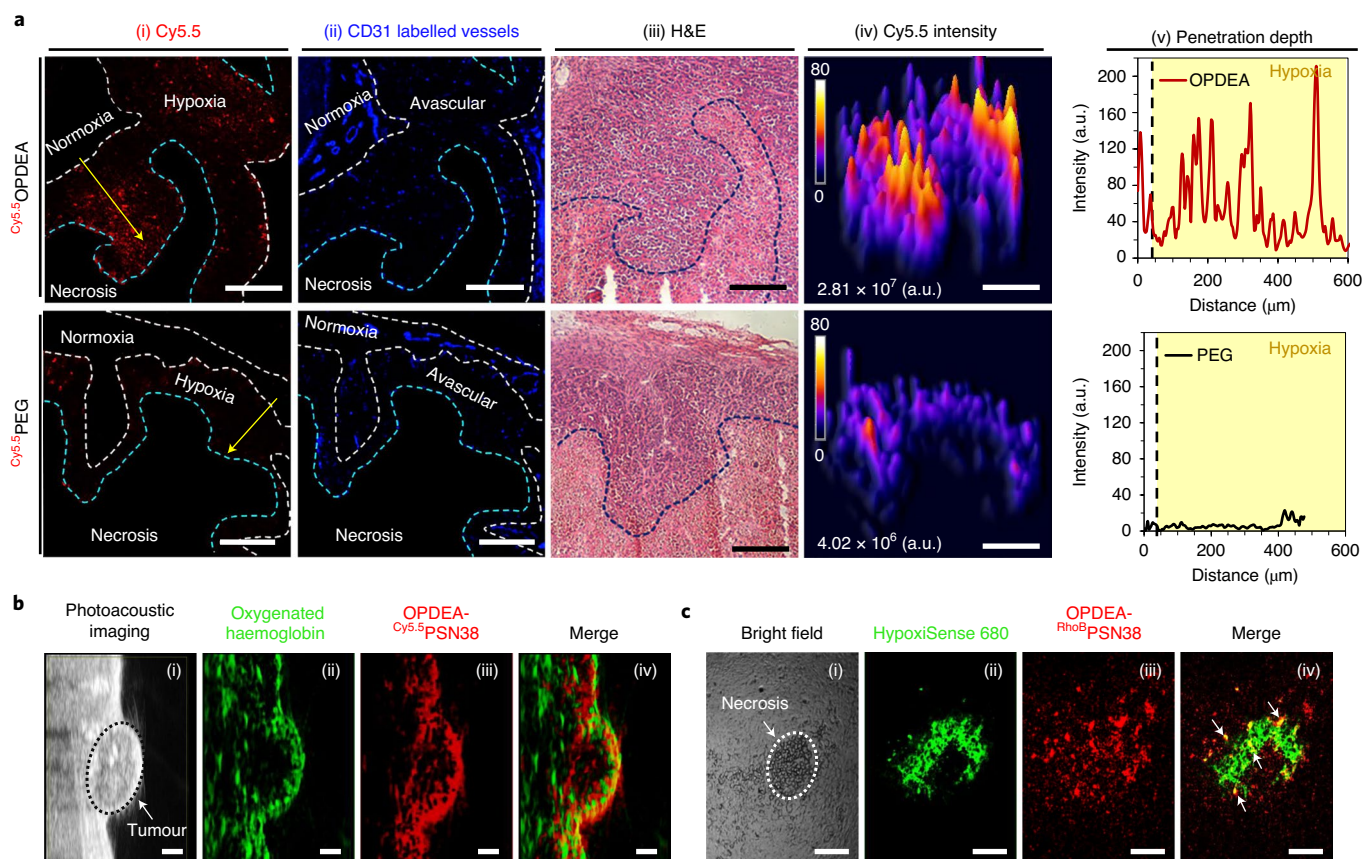


Fig. 5 | In vivo tumour penetration of OPDEA. **a**, Ex vivo analysis of the intratumoral distribution of Cy5.5-OPDEA or Cy5.5-PEG in a subcutaneous HepG2 tumour of about 10 mm^3 24 h after intravenous injection with a dose equivalent to 0.5 mg kg^{-1} Cy5.5 . Images show Cy5.5 fluorescence ((i); red), blood capillaries stained with anti-CD31 (blue) delineating vascular-rich, avascular and necrotic regions ((ii), H&E staining for identification of necrotic regions ((iii), an ImageJ 3D plot of Cy5.5 fluorescence density in the avascular hypoxic area ((iv), and fluorescence intensity along randomly selected lines from the normoxic to hypoxic regions (yellow arrows in (i)). Scale bars, $300 \mu\text{m}$. For each tumour, three adjacent sections were separately stained for analysis. Note that in the image of Cy5.5 fluorescence, the autofluorescence of the necrotic regions and Cy5.5 fluorescence of the normoxic regions were manually removed by image processing software^{46,47}; the original images are shown in Supplementary Fig. 26. **b**, In vivo photoacoustic images of the distribution of OPDEA- Cy5.5-PSN38 micelles (Cy5.5 -equivalent dose, 0.5 mg kg^{-1}) in a subcutaneous HepG2 tumour. Images shown are an ultrasound image of the tumour ((i); photoacoustic image of oxygenated haemoglobin ((ii); green); fluorescent image of OPDEA- Cy5.5-PSN38 (red) in the tumour ((iii); and the overlap of photoacoustic images of oxygenated haemoglobin (green) and OPDEA- Cy5.5-PSN38 micelles ((iv); red). Scale bars, 1.0 mm . Due to the limited availability of the photoacoustic imaging system, a single measurement was carried out for this experiment. **c**, Ex vivo fluorescent images showing the distribution of intravenously injected OPDEA- RhoB-PSN38 (RhoB -equivalent dose, 0.5 mg kg^{-1}) in hypoxic tumour regions (stained with HypoxiSense 680). $n = 3$ biological replicates. Images shown are a bright-field image of the tumour, with the necrotic area marked by a dotted line ((i); HypoxiSense 680 ((ii); green), OPDEA- RhoB-PSN38 ((iii); red) and an overlay, showing colocalization as yellow dots ((iv); marked by arrows). Scale bars, $100 \mu\text{m}$. A representative image of the PEG- Cy5.5-PSN38 control is shown in Supplementary Fig. 27.

five tumour-free mice remained healthy with no tumour recurrence, and the small tumour in the tumour-bearing mouse did not increase in size (Fig. 6c, images). The body weight of the mice in each group remained stable (Supplementary Fig. 31a).

The therapeutic effects of the treatments were analysed by H&E staining of tumours dissected on day 16 (Fig. 6d). Tumours from the PBS-treated group consisted of tightly packed cells with typical cancer cell morphology. The cells in the outer layer of the tumour treated with PEG-PSN38 were atrophic with intercellular-space dilatation and even small cavities, suggesting substantial damage and apoptosis. However, the interior cells had typical tumour cell morphology with few apoptotic cells. By contrast, the tumour treated with OPDEA-PSN38 micelles, similar to the OPDEA-PSN38-treated spheroids, contained fibre-like structures with large cavities, few tumour cells, low cellular integrity, disrupted and vague cytoplasm and condensed nuclei.

Large tumours are notoriously resistant to chemotherapy due to their exponential growth, poor drug penetration and hypoxia⁹. We further challenged OPDEA-PSN38 micelles by treating exponentially growing 500 mm^3 HepG2 tumours. We used irinotecan (CPT-11), a clinical first-line anticancer drug that is metabolized in vivo to SN38, as a positive control. Tumours in mice treated with PBS grew exponentially, and the mice had to be euthanized after 14 d due to the large tumour volumes (Fig. 6e, left). At an SN38-equivalent dose of 20 mg kg^{-1} , CPT-11 initially inhibited the tumour growth for about 8 d, after which they grew as quickly as those of the PBS control group, and the mice had to be euthanized on day 22. By contrast, the large tumours rapidly regressed following the first injection with OPDEA-PSN38 micelles at the same SN38-equivalent dose; the tumours continued to shrink after the second injection, were barely visible by day 20, and did not subsequently relapse (Fig. 6e, inset). The mice were euthanized on day 32 after the first

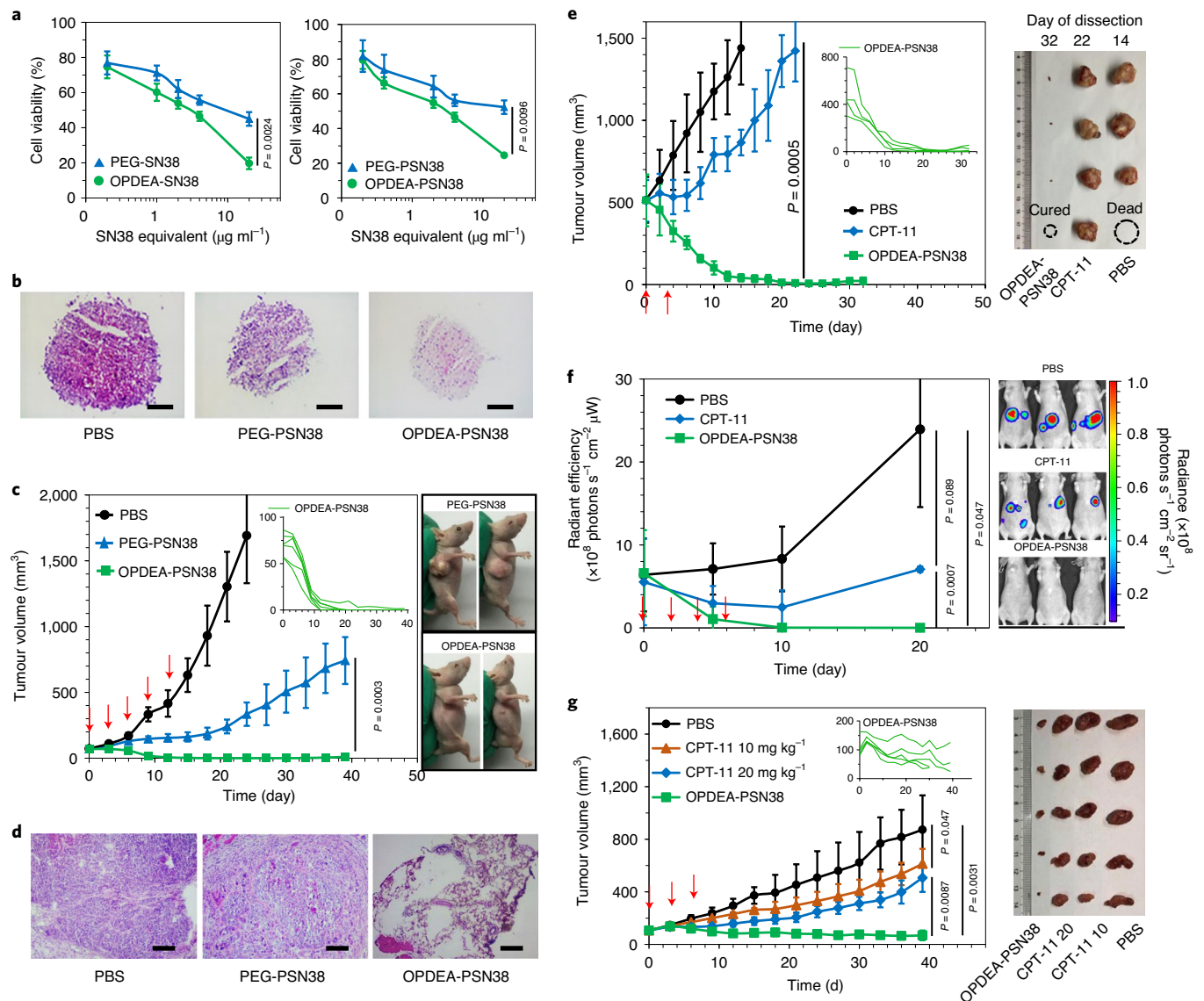


Fig. 6 | In vitro and in vivo anticancer activity of OPDEA conjugate and micelles. **a**, In vitro cytotoxicity of the OPDEA-SN38 conjugate and the OPDEA-PSN38 micelles against HepG2 MTSs determined by the resazurin assay; 96 h treatment. Data are mean $\pm$ s.d. of biological replicates ($n=3$); unpaired two-tailed Student's *t*-test. **b**, Representative images (from three images obtained) of the H&E-stained MTS sections after treatment with an SN38-equivalent dose of $20 \mu\text{g ml}^{-1}$ for 96 h. Scale bars, $200 \mu\text{m}$. The other images are shown in Supplementary Fig. 29. **c**, Anticancer activity of OPDEA-PSN38 micelles against subcutaneous HepG2 xenograft tumours. Left: average tumour growth curve of each group. Inset: individual tumour growth curves in the OPDEA-PSN38 group. Right: representative mice on day 60 after the first treatment. The five treatments were injected into the tail vein every 3 d with an SN38-equivalent dose of 10 mg kg^{-1} . **d**, Representative images (from three images obtained) of H&E-stained tumour sections from **c** 1 d after the final treatment. Scale bars, $100 \mu\text{m}$. **e**, Anticancer activity of OPDEA-PSN38 micelles against large subcutaneous HepG2 tumours compared with CPT-11. When the tumours reached about 500 mm^3 , the treatments were injected twice into the tail vein at an SN38-equivalent dose of 20 mg kg^{-1} . Left: average tumour growth curve of each group. Inset: individual tumour growth curves in the OPDEA-PSN38 group. Right: tumours dissected on day 14 for the PBS group, day 22 for the CPT-11 group, and day 32 for the OPDEA-PSN38 group. **f**, Antitumour activity of OPDEA-PSN38 micelles against orthotopic HepG2-Luci tumours illustrated by in vivo bioluminescence intensity curves (left), and images of tumour-bearing mice on day 20 (right). Treatments were injected into the tail vein as indicated by the arrows with an SN38-equivalent dose of 20 mg kg^{-1} . **g**, Antitumour activity of OPDEA-PSN38 micelles against a liver cancer PDX model. Left: average tumour growth curves. Inset: individual tumour growth curves in the OPDEA-PSN38 group. Right: images of dissected tumours. OPDEA-PSN38 micelles were injected into the tail vein every 2 d for 3 treatments at the indicated dose. **c, e–g**, Data are mean $\pm$ s.d. of biological replicates ($n=6$ (**c**), $n=4$ (**e**), $n=3$ (**f**) and $n=5$ (**g**)); one-way ANOVA with Tukey's correction and 95% confidence intervals.

treatment. One of the four OPDEA-PSN38-treated mice was entirely free of tumours, and the other three had small tumours of about 3 mm^3 (Fig. 6e). The body weight of the mice in all groups remained stable during and after the treatments (Supplementary Fig. 31b). Similarly, OPDEA-PSN38 micelles showed potent

antitumour activity against large (approximately 500 mm^3) HT29 tumours in the colorectal cancer model (Supplementary Fig. 32).

Next, we evaluated the anticancer activity of OPDEA-PSN38 against orthotopic tumours derived from hepatocellular carcinoma (Fig. 6f and Supplementary Fig. 33), which better mimic the tumour

growth microenvironment and are more clinically relevant. The orthotopic tumours were established by implanting tumour fragments comprising HepG2 cells stably expressing firefly luciferase (HepG2-Luci) into the liver parenchyma of nude mice. After 7 d, tumour formation was confirmed by euthanizing one of the mice, and tumour growth was monitored by *in vivo* bioluminescence imaging. Tumours in the PBS control group grew rapidly. CPT-11 inhibited the growth of the orthotopic tumours, but they relapsed after withdrawal of the treatment. By contrast, four treatments with OPDEA-PSN38 micelles at an SN38-equivalent dose of 20 mg kg^{-1} rapidly reduced the size of the tumours and eliminated them within 10 d after the first treatment. No sign of recurrence was observed during the treatment-free period.

Patient-derived xenografts (PDX) have similar histological and pathological characteristics and similar responses to anticancer agents as human tumours⁵⁰, and provide a more reliable prediction of the clinical response to therapeutic agents⁵¹. Thus, we further tested OPDEA-PSN38 micelles on the PDX model (hepatocellular carcinoma, P51F3; see Supplementary Table 2 for detailed information). Similar to the cell-derived models, three intravenous injections of OPDEA-PSN38 micelles also gradually shrank the PDX tumours at an SN38-equivalent dose of 10 mg kg^{-1} , whereas CPT-11 at SN38-equivalent doses of 10 or even 20 mg kg^{-1} only slightly inhibited the tumour growth (Fig. 6g). On day 39, the PDX tumours of the OPDEA-PSN38 treated mice were significantly smaller than those from the PBS group (mean tumour volumes of approximately 68.9 mm^3 versus 872.6 mm^3) (Fig. 6g, right); the body weights of mice in these groups remained stable (Supplementary Fig. 34). Tests in other PDX models (P25F3) showed similar results (Supplementary Fig. 35). These PDX results not only validated the potency of OPDEA-PSN38 micelles but also indicated that OPDEA-PSN38 would be effective against human tumours in the clinic.

Liposomal irinotecan (Onivyde) was recently approved by the US Food and Drug Administration and has shown substantial clinical potential⁵⁶. We compared the therapeutic effects of OPDEA-PSN38 with a liposomal CPT-11 formulation (Lipo-CPT-11) similar to that of Onivyde (Supplementary Fig. 36). Lipo-CPT-11 and OPDEA-PSN38 both inhibited the HT29 tumour growth, but tumours treated with Lipo-CPT-11 relapsed after withdrawal of the treatment. By contrast, tumours treated with OPDEA-PSN38 continued to shrink—two were completely eradicated, and the remaining tumours did not subsequently relapse.

Mechanism of effective drug delivery by OPDEA. As discussed above, OPDEA is polyzwitterionic and thus antifouling towards proteins (Supplementary Fig. 37), similar to other *N*-oxide polymers⁵² and polyzwitterions^{16,41,53}, but could bind cells. Simulation using the Becke's half-and-half functional (BHandH) with the (6-311++G(d,p)) basis sets (BHandH/6-311++G(d,p))⁵⁴ and the Minnesota 06 density functional (M06-2X) with the (6-311++G(d,p)) basis sets (M06-2X/6-311++G(d,p))⁵⁵ methods showed that the *N*-oxide moiety interacted via multiple weak hydrogen bonds and charges with PC or PE with the binding energies estimated to be about -22 to $-16 \text{ kcal mol}^{-1}$ (Fig. 7a). Isothermal titration calorimetry (ITC) experiments showed that PC and PE bound OPDEA exothermically, and the calculated binding constant K_a was 38.5 mol^{-1} for PC and 21.8 mol^{-1} for PE (Supplementary Fig. 38a,b). The multiple bonds between OPDEA and the abundant PC and PE in the cell membrane would produce a cooperative effect and enable its adsorption on the cell membrane. This was validated by surface plasmon resonance⁵⁶ (SPR) (Supplementary Fig. 38c), in which OPDEA in a flow quickly adsorbed onto a PC-coated surface and, to a lesser extent, on a PE-coated surface, whereas PEG did not bind to either surface, as further visualized by fluorescent imaging (Supplementary Fig. 38d). All cell membranes are largely composed of PC and PE⁵⁷. The binding constant of OPDEA with cell membranes

measured by ITC was $4.26 \times 10^4 \text{ mol}^{-1}$ for RBCs, $1.56 \times 10^4 \text{ mol}^{-1}$ for immortalized human umbilical vein endothelial cells (HUVECs), $6.83 \times 10^4 \text{ mol}^{-1}$ for hepatocellular carcinoma-derived endothelial (ECDHCC-1) cells and $1.40 \times 10^5 \text{ mol}^{-1}$ for HepG2 tumour cells (Supplementary Fig. 38e–i). Thus, the zwitterionic poly(*N*-oxide) is atypical in that it combines non-stickiness towards proteins with binding of phospholipid heads in contrast to cell-resistant polyzwitterions^{53,58,59}.

Adsorption of OPDEA onto RBCs was further examined and compared with PEG and two well-known polyzwitterions: poly(choline phosphate)³⁶ (PolyCP), which was non-sticky towards proteins but bound strongly to RBCs (Supplementary Fig. 39), and poly(methacryloyloxyethyl phosphorylcholine)⁶⁰ (PolyPC), which was non-fouling to proteins or cells (Supplementary Fig. 40). As expected, RBCs incubated with Cy5.5-PEG or Cy5.5-PolyPC exhibited little fluorescence, whereas RBCs incubated with Cy5.5-PolyCP or Cy5.5-OPDEA exhibited strong Cy5.5 fluorescence (Fig. 7b). The two RBC-binding polymers differed in that Cy5.5-PolyCP caused RBCs to aggregate and become non-dispersible in the erythrocyte buffer containing 0.5 mg ml^{-1} heparin (Fig. 7b and Supplementary Fig. 41), whereas Cy5.5-OPDEA adsorption did not markedly change the properties of the RBCs, including dispersibility, cell integrity and morphology (Fig. 7b and Supplementary Figs. 19, 41 and 42). The adsorbed Cy5.5-OPDEA on RBCs was easily removed by manual extrusion mimicking the shear stress in the circulatory system (Fig. 7c and Supplementary Figs. 41 and 42). Linescan analysis showed good overlap between fluorescence of Cy5.5-OPDEA and that of RBC membranes stained with wheat germ agglutinin (WGA) labelled with fluorescein isothiocyanate (FITC) (FITC-WGA) (Fig. 7d), indicating that OPDEA adhered to the surface of RBCs without entering their cytosol and thus easily fell off, in contrast to hydrophobic polystyrene nanoparticles⁶¹ and PolyCP, which adsorbed more strongly and altered the cells' properties.

The desorption kinetics of OPDEA- Cy5.5-PSN38 from RBCs was probed using flow cytometry. RBCs were incubated with OPDEA- Cy5.5-PSN38 for 1 h, and RBCs adsorbing OPDEA- Cy5.5-PSN38 (OPDEA- Cy5.5-PSN38@RBCs) were isolated. Upon dispersion in erythrocyte buffer, nearly 45% of OPDEA- Cy5.5-PSN38 desorbed from RBCs within 5 min, reaching equilibrium within 60 min (Supplementary Fig. 43a). Thus, the adsorption–desorption of OPDEA- Cy5.5-PSN38 on RBCs was a fast dynamic event.

Following intravenous injection, OPDEA or OPDEA micelles immediately dispersed among the plasma and RBCs. Ten minutes after injection, about 4.25% of Cy5.5-OPDEA and 7.18% of OPDEA- Cy5.5-PSN38 micelles in the blood were adsorbed on RBCs; these fractions increased to 23.18% and 14.63%, respectively, at 2 h after injection (Fig. 7e,f). Thus, unlike polystyrene nanoparticles^{32,62}, hitchhiking RBCs only partly contributed to the long lifetime of OPDEA and its micelles in the circulation.

Thus, the weak binding to RBCs and non-stickiness towards proteins of OPDEA—in contrast to polystyrene^{32,62} and PolyCP³⁶—enabled OPDEA or OPDEA micelles to circulate either in the plasma or to reversibly hitchhike on RBCs, thus providing them an opportunity to bind to the tumour luminal endothelial surface (Fig. 1b(i)). We next evaluated the competition for OPDEA- Cy5.5-PSN38 micelles between RBCs and HUVECs or ECDHCC-1 cells. We co-cultured RBCs or endothelial cells with OPDEA- Cy5.5-PSN38 for 5, 30 and 60 min, then isolated the cells for fluorescence intensity analysis using flow cytometry. Some OPDEA- Cy5.5-PSN38 micelles were adsorbed on RBCs, but the amount of adsorption did not increase with time; by contrast, a larger amount of OPDEA- Cy5.5-PSN38 micelles was associated with endothelial cells within 5 min, and this amount increased with time (Supplementary Fig. 43b). We then tested the ability of OPDEA- Cy5.5-PSN38 micelles bound on RBCs to translocate to tECs. RBCs adsorbing OPDEA- Cy5.5-PSN38 micelles (OPDEA- Cy5.5-PSN38@RBCs) were isolated and co-cultured with

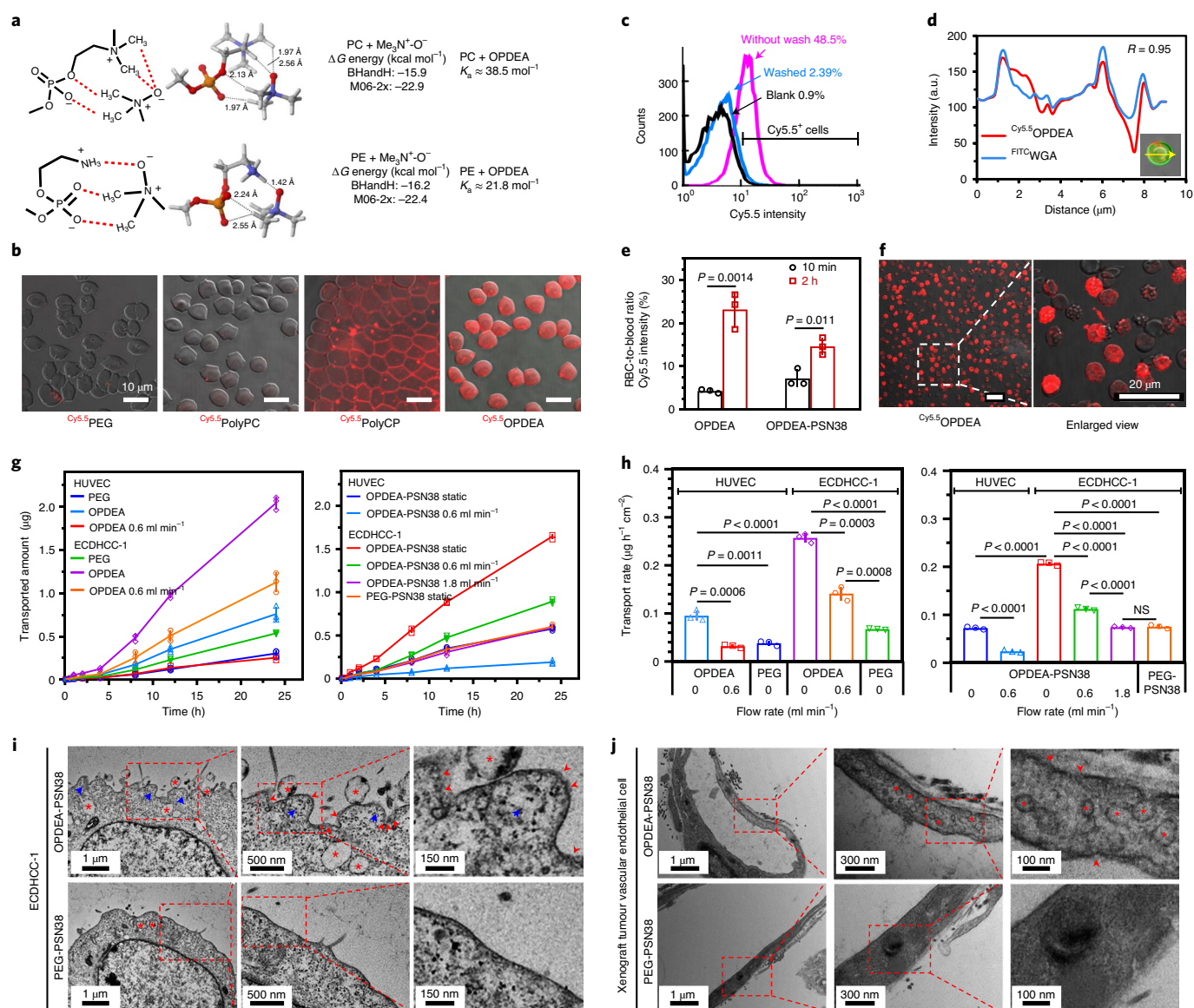


Fig. 7 | Mechanistic studies of RBC binding- and active transcytosis-based trans-endothelial transportation of OPDEA. **a**, Binding-energy simulation report for an *N*-oxide-trimethylamine moiety with PC or PE ionic moieties, and binding constants (K_a) of OPDEA with PC or PE calculated from the ITC experiments (Supplementary Fig. 38). $n = 3$ biological replicates. **b**, Representative CLSM images of RBCs adsorbing Cy5.5-labelled polymers (Cy5.5-equivalent dose, $0.5 \mu\text{g ml}^{-1}$; 3 h incubation), out of 3 images obtained. **c**, Cy5.5-positive RBCs, detected by flow cytometry of Cy5.5OPDEA -treated RBCs with or without washing (Cy5.5-equivalent dose, $0.5 \mu\text{g ml}^{-1}$; 1 h incubation). RBCs were washed by extrusion through a syringe. Data are expressed as the median of 3 biological replicates; 10,000 cells analysed per group. **d**, Colocalization of FITC-WGA-stained RBC membrane (blue line) and Cy5.5OPDEA (red line) along the selected direction (yellow arrow) in CLSM images of a randomly selected RBC (out of 3 RBCs analysed; Cy5.5-equivalent dose, $0.5 \mu\text{g ml}^{-1}$; 1 h incubation). The other images are shown in Supplementary Fig. 42. **e**, The ratio of Cy5.5OPDEA or OPDEA-Cy5.5PSN38 bound on RBCs to that in blood 10 min or 2 h after tail-vein injection (Cy5.5-equivalent dose, 0.5 mg kg^{-1}). Fluorescence intensities of $50 \mu\text{l}$ of whole blood and the isolated RBCs were determined using the IVIS Spectrum imaging system. **f**, Fluorescence images of RBCs 2 min after tail-vein injection of Cy5.5OPDEA (Cy5.5-equivalent dose, 0.5 mg kg^{-1}). $n = 3$ biological replicates. **g, h**, Trans-endothelial transport of Cy5.5-labelled materials across the endothelial cell monolayer on a transwell membrane, measured by cumulative transport amount (**g**) and rate (**h**). Endothelial cells were incubated with Cy5.5OPDEA or Cy5.5PEG (Cy5.5-equivalent dose, $0.8 \mu\text{g ml}^{-1}$), or OPDEA-Cy5.5PSN38 or PEG-Cy5.5PSN38 (Cy5.5-equivalent dose, $0.4 \mu\text{g ml}^{-1}$) under static or flow conditions (0.6 or 1.8 ml min^{-1}). **i, j**, Representative transmission electron microscopy images of the ultrastructure of a ECDHCC-1 cell (**i**) and subcutaneous HepG2 xenograft tumour blood vessel (**j**) after treatment with OPDEA-PSN38 or PEG-PSN38 micelles ($n = 3$ biological replicates). Microvesicles (red asterisks), pseudopods (blue arrowheads) and fenestrae (red arrowheads) are indicated. Additional images are shown in Supplementary Figs. 50 and 51. **e, g, h**, Data are mean $\pm$ s.d. of biological replicates ($n = 3$); one-way ANOVA with Tukey's correction and 95% confidence intervals.

ECDHCC-1 cells under flow conditions. After 30 min, a large amount of OPDEA-Cy5.5PSN38 micelles was transferred to ECDHCC-1 cells, and this amount increased with time (Supplementary Fig. 43c). A trans-endothelial cell-layer transport experiment showed that

while the amount of micelles bound on RBCs decreased with time (Supplementary Fig. 43d), the micelle content in the medium of the basolateral compartment increased (Supplementary Fig. 43e), indicating that OPDEA-Cy5.5PSN38 continuously translocated

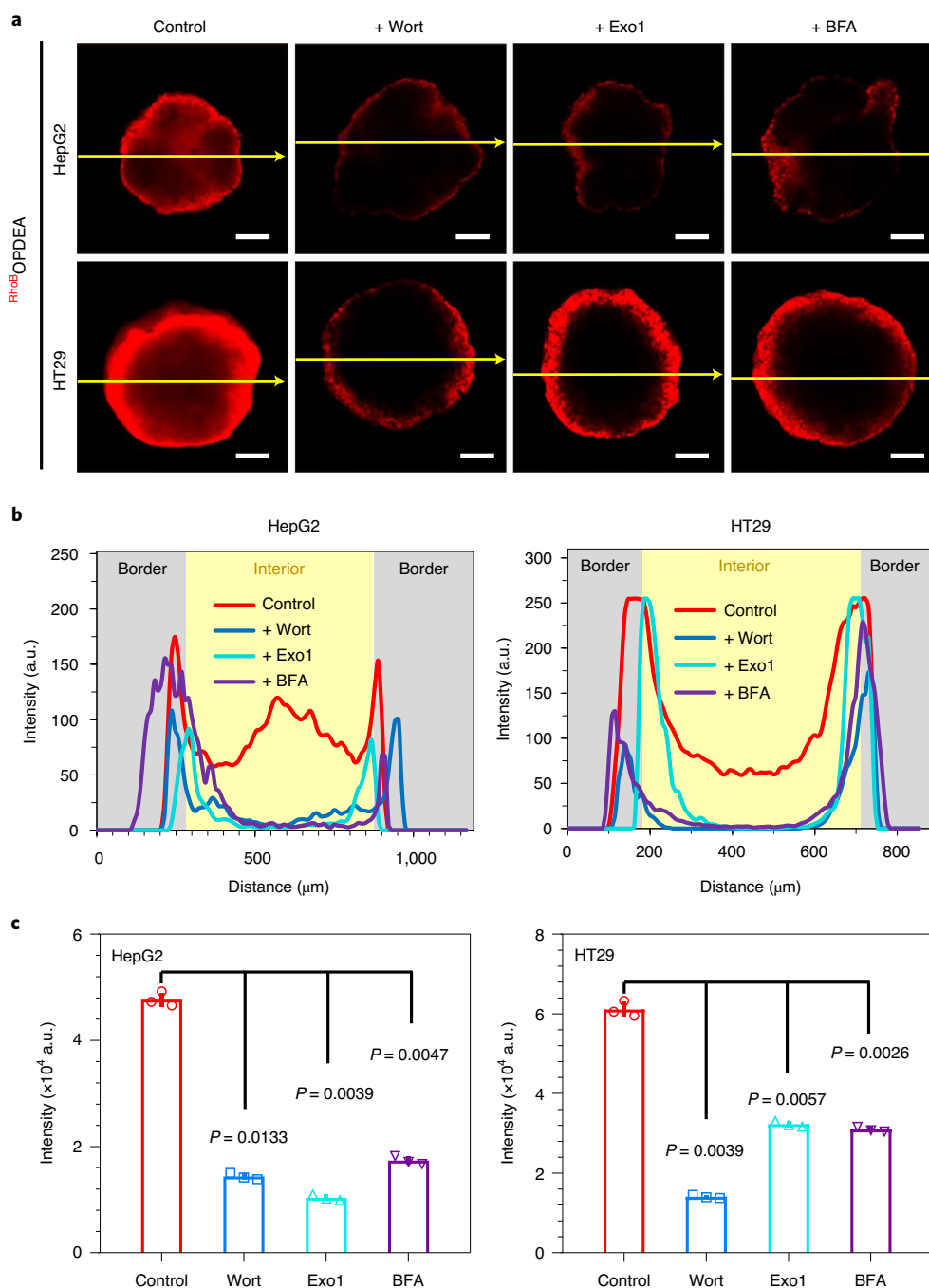


Fig. 8 | The effects of inhibitors of endo- and exocytosis on the penetration of $\text{RhoB}^{\text{OPDEA}}$ in HepG2 or HT29 spheroids. **a–c, MTSs were separately pre-treated with the inhibitors for 10 h and $\text{RhoB}^{\text{OPDEA}}$ (RhoB-equivalent dose, $5 \mu\text{g ml}^{-1}$) for 4 h, and then imaged with CLSM by Z-stack tomoscan at $20 \mu\text{m}$ intervals. **a**, Representative images of $\text{RhoB}^{\text{OPDEA}}$ distribution in the middle layers of spheroids, out of five images obtained per group. Scale bars, $200 \mu\text{m}$. Additional images are shown in Supplementary Fig. 52. BFA, brefeldin A; Wort, wortmannin. **b**, $\text{RhoB}^{\text{OPDEA}}$ fluorescence intensity along the yellow arrows in **a**. **c**, Integration of total fluorescence of the layers in **a**. Data are mean $\pm$ s.d. of biological replicates ($n=3$); one-way ANOVA with Tukey's correction and 95% confidence intervals.**

from RBCs to tECs for endocytosis and subsequent transcytosis. HUVECs and ECDHCC-1 cells internalized OPDEA-Cy5.5PSN38 micelles or Cy5.5OPDEA efficiently (Supplementary Fig. 44). Thus OPDEA-Cy5.5PSN38 or Cy5.5OPDEA could adhere to both RBCs and endothelial cells, but those bound to endothelial cells could trigger endocytosis and stimulate internalization.

So, while adsorption on the luminal surface facilitated extravasation via the EPR effect³⁸ through the leaky tumour vasculature,

including the widespread endothelial fenestrae, transcellular holes and large intercellular openings⁶³ (Fig. 1b(ii)), adsorption on the cell surface may also induce transcytosis of tECs³⁰, which were recently considered responsible for the extravasation of nanomaterials into tumour interstitium instead of the EPR effect³³.

We first estimated trans-endothelial transport of Cy5.5OPDEA in vitro by measuring its rates crossing the monolayer of HUVECs or ECDHCC-1 cells on the transwell membrane (Fig. 7g,h). Cy5.5PEG ,

very little of which crossed both cell layers, was used as a reference. In static conditions, $Cy5.5$ OPDEA crossed the tumour endothelial ECDHCC-1 monolayer at a rate of $0.26 \mu\text{g h}^{-1} \text{cm}^{-2}$, a much faster rate than through the HUVEC monolayer ($0.10 \mu\text{g h}^{-1} \text{cm}^{-2}$). Adding a flow (of around 0.6 ml min^{-1} or 0.5 mm s^{-1}) slowed the trans-endothelial transport by 1.8–3.0 times, particularly through the normal HUVEC layer. Flow cytometry showed that $Cy5.5$ OPDEA was internalized by both cell types, but ECDHCC-1 took up about twice as much $Cy5.5$ OPDEA compared with HUVECs (Supplementary Fig. 45), indicating that tECs were more actively endocytotic than normal endothelial cells. $Cy5.5$ OPDEA was also taken up via macropinocytosis (Supplementary Fig. 46), mostly into endoplasmic reticulum and Golgi apparatus (Supplementary Fig. 47), where materials were packaged for exocytosis⁶⁴. Furthermore, transport across the ECDHCC-1 layer was unaffected by nocodazole (a microtubule inhibitor), but was slowed by a macropinocytosis inhibitor (cytochalasin D), Golgi disruption (monensin) or blocking of protein transport to the Golgi (brefeldin A) (Supplementary Fig. 48), indicating that the transport occurred through transcytosis of OPDEA or OPDEA micelles through the Golgi apparatus.

Next, we investigated *in vivo* the process by which $Cy5.5$ OPDEA dissociated from RBCs and bound tECs, triggering extravasation and penetration into a tumour (Supplementary Fig. 49a). $Cy5.5$ OPDEA was mixed with RBCs, and RBCs adsorbing $Cy5.5$ OPDEA ($Cy5.5$ OPDEA@RBCs) were isolated. The $Cy5.5$ OPDEA@RBCs injected intravenously resulted in extravasation of $Cy5.5$ OPDEA into the tumour tissue, confirming that $Cy5.5$ OPDEA translocated from $Cy5.5$ OPDEA@RBCs to the tECs and extravasated into the tumour. Importantly, intratumoral administration of an exocytosis inhibitor, Exo1, abolished the extravasation of $Cy5.5$ OPDEA (Supplementary Fig. 49b), confirming transcytosis-based extravasation.

Imaging the cross-section of ECDHCC-1 cells treated with OPDEA-PSN38 micelles showed that these cells contained vesicles (indicated by red asterisks, relating to endo- and exocytosis), pseudopods (indicated by blue arrows, relating to macropinocytosis), and fenestrae (indicated by red arrows, relating to transcytosis)^{63,65}, but they were not found in cells treated with PEG-PSN38 micelles (Fig. 7i and Supplementary Fig. 50). These phenomena were also observed *in vivo*. The tECs of the blood capillaries in the OPDEA-PSN38-treated tumours contained many vesicles, whereas such vesicles were rare in the tECs of the blood capillaries in PEG-PSN38-treated tumours (Fig. 7j and Supplementary Fig. 51). Together, these results showed that OPDEA or OPDEA micelles can adhere to tECs and trigger transcytosis.

Endothelial cells of blood capillaries in healthy tissues are less active than the immortalized HUVEC cells. Furthermore, the average speed of blood flow is $4.9\text{--}19 \text{ cm s}^{-1}$ in arteries and $1.5\text{--}7.1 \text{ cm s}^{-1}$ in veins⁶⁶ in healthy tissues. Thus, $Cy5.5$ OPDEA that weakly attached to the surface of healthy endothelial cells may be flushed away before triggering endocytosis because, as shown in Fig. 7g, even adding a nominal flow (about 0.5 mm s^{-1}) reduced the trans-endothelial transport substantially. Blood flows in tumour capillaries are slow and sometimes stagnant⁶⁷. $Cy5.5$ OPDEA adsorbing on the more active tECs triggered fast endocytosis and subsequent transcytosis, accounting for the tumour-selective extravasation seen in Supplementary Fig. 25. Furthermore, by comparison with the slow extravasation of $Cy5.5$ PEG (Fig. 4b), the faster extravasation of OPDEA indicated that AMT (Fig. 1b(iii)) may contribute more than the EPR effect to OPDEA extravasation (Fig. 1b(ii)), consistent with a recent report³³.

The optimal cell affinity of OPDEA is critical: if the cell affinity was too strong, OPDEA would adsorb on RBCs unable to translocate to the endothelium for extravasation, but would require mechanical transfer in the microvasculature and would end up mostly in the lung⁶⁸ or the first organ downstream of the injection site³². The quantitative correlations among cell affinity, blood circulation and tumour accumulation require further study.

Another important property of OPDEA and OPDEA micelles is their tumour-penetration ability (Fig. 5). Infiltration of nanomedicine into tumour occurs either by paracellular passive diffusion through the extracellular matrix or by intercellular delivery among tumour cells. We used MTSs to probe the involvement of cancer cells in OPDEA penetration into tumours. As OPDEA and its micelles were endocytosed through the macropinocytosis-related pathway, we used wortmannin to inhibit cell endocytosis. Exo1 and brefeldin A, which inhibit exocytosis by causing a collapse of the Golgi apparatus⁶⁹, were used to disable the exocytosis. Figure 8a and Supplementary Fig. 52 show that pre-treatment of MTSs with wortmannin, Exo1 or brefeldin A abolished the penetration of $RhoB$ OPDEA and confined the fluorescence to the outer layers of both types of MTSs. Quantitative analysis of the fluorescence signal showed that the inhibitor-treated MTSs had substantially reduced $RhoB$ OPDEA in the centre compared to control MTSs (Fig. 8b). As a result, the overall content of $RhoB$ OPDEA in the entire MTS layer was significantly reduced (Fig. 8c). Consequently, adding the endocytosis inhibitor (wortmannin) or exocytosis inhibitor (brefeldin A) decreased the cytotoxicity of OPDEA-PSN38 to the MTSs (Supplementary Fig. 53). For comparison, the $Cy5.5$ PolyCP adsorbed too strongly on the cell membrane to trigger endo- and exocytosis, and it remained on the surface of the MTSs (Supplementary Fig. 54). These results indicated that the penetration and accumulation of OPDEA in MTSs relied on proper cell-binding-triggered transcytosis in cancer cells.

An ‘infection’ experiment further confirmed the intercellular delivery of $RhoB$ OPDEA via endo- and exocytosis. We preincubated HepG2 or HT29 cells with $RhoB$ OPDEA in glass-bottom dishes for 5 h. These cells took up a substantial amount of $RhoB$ OPDEA (Supplementary Fig. 55(i)). We washed and then cultured these cells in fresh medium for 12 h and found that the fluorescence intensity in the cells decreased (Supplementary Fig. 55(ii)) while $RhoB$ fluorescence was observed in the medium, and new cells cultured in this medium gained strong fluorescence (Supplementary Fig. 55(iii)). $RhoB$ OPDEA could therefore be passed to different batches of cells (Supplementary Fig. 55(iv)). Thus, the cells that internalized $RhoB$ OPDEA released some of it into the medium, which was subsequently taken up by other cells, that is, endo- and exocytosis (transcytosis). The transcytosis ability of OPDEA was dependent on the fast nondigestive macropinocytosis pathway. Macropinosomes are known to be leaky^{70,71} and can release OPDEA and OPDEA micelles into the cytosol, enabling translocation to the Golgi apparatus for exocytosis.

Together, these results indicated that the penetration of OPDEA and its micelles in tumours did not occur via paracellular diffusion but rather through an intercellular delivery process: the cells took up OPDEA from the perivascular side and exocytosed some to the other side, finally passing it on to avascular regions (Fig. 1b(iv)). This cancer cell transcytosis actively transferred OPDEA and circumvented the tumour’s intrinsic physiological barriers to paracellular diffusion, including high cell and matrix density and high osmotic pressure^{9,12}. This active tumour penetration enabled the nanomedicine to reach distal cells, even in hypoxic regions (Fig. 5).

Limited tumour penetration is considered one of the main drug-resistance mechanisms of the tumour microenvironment, accounting for the limited therapeutic efficacy of drugs and nanomedicines towards solid tumours⁷². As shown by cytotoxicity to MTSs, PEG-PSN38 could only kill the outer layers of cells on the MTS, and the cell in the interior remained undamaged. By contrast, OPDEA-PSN38 micelles simultaneously killed both outer and interior cells (Fig. 6b). Thus, the OPDEA-PSN38 micelles’ superior tumour-penetration ability enabled them to reach distal cells and release active SN38 intracellularly, leading to high therapeutic efficacy in eradicating small and exponentially growing large tumours (Fig. 6).

In summary, distinct from the strategy of using complex structures to integrate different functions for effective cancer

drug delivery, the non-stickiness towards proteins and weak phospholipid-binding properties of OPDEA provide useful characteristics for efficient cancer drug delivery; that is, prolonged circulation in blood plasma, hitchhiking on RBCs, effective tumour accumulation, adhesion to tECs for facilitated extravasation into tumours mainly through transcytosis, adsorption on cancer cells to trigger fast cellular uptake for intracellular drug release and also transcytosis-based active tumour penetration. These abilities make OPDEA-based nanomedicines therapeutically potent. Moreover, because of the chemical simplicity of the *N*-oxide moiety, OPDEA-anticancer-drug conjugates may offer translational advantages.

Methods

Cell lines and mice. HepG2, A549 (human lung carcinoma), HeLa (human cervix adenocarcinoma), HT29, HUVECs and 4T1-GFP (mouse mammary carcinoma expressing green fluorescent protein) cell lines were obtained from the Cell Bank of the Chinese Academy of Sciences. The ECDHCC-1 cell line was purchased from Fusheng Industrial. HepG2-Luci cells were established by lentivirus transfection of the firefly luciferase gene (Ubi-MCS-SV40-firefly luciferase-IRES-puromycin) into HepG2 cells according to the manufacturer's protocol (Shanghai Genechem). A549, HeLa and HT29 cell lines were incubated in a nutritious 1640 medium (containing 10% FBS and 1% (v/v) penicillin-streptomycin) at 37 °C with 5% CO₂; HepG2, HepG2-Luci, HUVEC and ECDHCC-1 cells were incubated in nutritious DMEM medium.

BALB/c-nude mice and female ICR mice (female, 6–8 weeks old, 18 g) were purchased from Shanghai Slacces Laboratory Animal. NOD/SCID mice (male, 4 weeks old, 18 g) were purchased from the Model Animal Research Center of Nanjing University. Mice were housed in the Zhejiang Academy of Medical Sciences under specific pathogen-reduced conditions.

Animal and human use statement. All animal experiments were carried out according to the protocols approved by the Zhejiang Academy of Medical Sciences and UK Co-ordinating Committee on Cancer Research guidelines. Surgically resected human liver cancer tissues were obtained from patients with liver cancer with identifiers P25 and P51 following the protocol approved by the First Affiliated Hospital of Zhejiang University School of Medicine.

Synthesis of PDEA. CuBr (0.033 g, 0.23 mmol) and 2,2'-bipyridine (0.071 g, 0.45 mmol) were added to a Schlenk flask and degassed by five repeated freeze-pump-thaw cycles with nitrogen. The degassed solution of DEA (5.08 g, 27.4 mmol) in methanol (2.0 ml) was injected into the flask under a nitrogen atmosphere. Ethyl α -bromoisobutyrate (0.033 g, 0.17 mmol) was injected into the flask through a microsyringe. The solution was stirred at 30 °C for 4 h. The polymerization was quenched by diluting with 100 ml THF. The mixture was passed through an alumina column using THF as the eluent, concentrated, and poured into cold *n*-hexane. The resulting PDEA (2.7 g) was isolated and then dried under vacuum. ¹H-NMR spectrum (400 MHz, CDCl₃) δ ppm: 3.9–4.1 (2H, COOCH₂), 2.6–2.8 (2H, COOCH₂CH₂N), 2.5–2.6 (4H, N(CH₂CH₃)₂), 1.7–2.1 (2H, CH₂CH₃), 0.8–1.1 (9H, N(CH₂CH₃)₂, CH₃CCOOCH₂). The number-averaged molecular mass (*M_n*) of the polymer was determined by gel permeation chromatography (GPC) to be 2.1×10^4 with a PDI of 1.2.

Synthesis of poly(DEA-co-*tert*-butyl methacrylate). The random copolymer of DEA with *tert*-butyl methacrylate (*t*BuMMA) was synthesized similarly to that of PDEA except that the monomer was a mixture of DEA (5.26 g, 28.4 mmol) and *t*BuMMA (0.28 g, 1.99 mmol) in 2.5 ml methanol. The molar ratio of *t*BuMMA in the polymer was 7 mol% of the repeating units, determined by the ¹H-NMR spectrum. ¹H-NMR spectrum (400 MHz, CDCl₃) δ ppm: 3.9–4.1 (1.8H), 2.5–2.8 (5.6H), 1.7–2.1 (1.9H), 1.3–1.5 (0.8H), 0.8–1.1 (8.5H). The number-averaged molecular mass (*M_n*) of the PDEA/*t*BuMMA, determined by GPC using THF as the mobile phase, was 2.2×10^4 with a PDI of 1.3.

Oxidation of the PDEA to OPDEA or OPDEA/*t*BuMMA. PDEA (0.5 g) was dissolved in 4.0 ml of 30% H₂O₂ solution. The mixture was stirred at room temperature for 4 h and then dialysed against deionized water (MWCO: 3,500 Da) to remove the unreacted H₂O₂. The polymer OPDEA (0.498 g, 92% yield) was obtained by lyophilization. The oxidation of PDEA/*t*BuMMA using the same procedure produced the *N*-oxide copolymer OPDEA/*t*BuMMA at a 94% yield. ¹H-NMR spectrum (400 MHz, D₂O) δ ppm: 4.1–4.5 (1.8H), 3.1–3.6 (5.7H), 1.7–2.1 (1.7H), 0.7–1.4 (9.6H).

Synthesis of poly(2-(*N*-oxide-*N,N*-diethylamino)ethyl methacrylate-co-methacrylic acid) and its *N*-hydroxysuccinimide ester. OPDEA/*t*BuMMA (3.31 g) was dissolved in 20 ml of CH₂Cl₂ and 10 ml of trifluoroacetic acid (TFA). This solution was stirred for 6 h at room temperature.

The solvent and TFA were removed by rotary evaporation, and the solid was dissolved in methanol. The solution was dialysed against methanol (MWCO: 3,500 Da) for 24 h and then against deionized water overnight. The carboxylic acid-functionalized polymer OPDEA/COOH (2.7 g) was obtained after lyophilization. The ¹H-NMR spectrum shows that all the *t*BuMMA units were hydrolyzed, and thus, the OPDEA/COOH contained 7% methacrylic acid monomer units. ¹H-NMR spectrum (400 MHz, D₂O) δ ppm: 4.1–4.5 (1.8H), 3.1–3.6 (5.6H), 1.7–2.1 (1.6H), 0.7–1.4 (9.0H).

The obtained OPDEA/COOH (0.3317 g, 0.12 mmol COOH), EDC-HCl (0.104 g, 0.547 mmol), NHS (0.0737 g, 0.640 mmol), a catalytic amount of DMAP, and 0.3 ml triethylamine (TEA) were dissolved in 5 ml anhydrous DMSO. The solution was stirred overnight at room temperature and precipitated in THF. The isolated solid was dissolved in 1.0 ml DMSO and dialysed against anhydrous DMSO (MWCO: 3,500 Da) for 6 h. OPDEA/COONHS was obtained by lyophilization (0.27 g). The NHS ester content in the polymer was 4%. ¹H-NMR spectrum (400 MHz, D₂O) δ ppm: 4.1–4.5 (1.8H), 3.1–3.6 (5.6H), 2.7–2.9 (0.16H), 1.7–2.1 (1.7H), 0.7–1.4 (9.0H).

Synthesis of OPDEA-SN38. OPDEA/COONHS (0.65 g, 0.28 mmol NHS ester), SN38-NH₂ (0.15 g, 0.32 mmol) and 0.2 ml TEA were dissolved in 5 ml anhydrous DMSO. The solution was stirred at room temperature for 6 h. The resulting solution was dialysed against DMSO (MWCO: 3,500 Da) overnight until no free SN38 was detected in the dialysate. The solution was lyophilized and the OPDEA-SN38 was obtained (0.66 g, 88% yield). The SN38-content was determined by high-performance liquid chromatography (HPLC) to be 3.7 wt%.

Synthesis of PEG-SN38 and PEG-PSN38. mPEG_{20k}-SPA (0.4691 g, 23.4 μ mol), SN38-NH₂ (20.4 mg, 44.0 μ mol), EDC-HCl (10.4 mg, 54.7 μ mol), and 0.5 ml Et₃N were dissolved in 6 ml anhydrous DMSO. The solution was stirred at room temperature for 24 h and dialysed against DMSO and then against deionized water (MWCO: 5.0 kDa). PEG-SN38 was obtained by lyophilization (0.38 g, 81% yield). The SN38 content was 1.9 wt% measured by HPLC.

PEG-PSN38 was synthesized via ATRP and characterized as described⁴⁰.

Synthesis of OPDEA-PSN38. The block copolymer of PDEA-PSN38 was prepared using ATRP, as reported previously⁴⁰. In brief, SN38-MA (0.2 g, 0.37 mmol), CuBr (1.0 mg, 7.0 μ mol) and *N,N,N',N',N''*-pentamethyldiethylenetriamine (40 μ l, 0.18 mmol) were dissolved in 1 ml of DMF. The solution was degassed with nitrogen for 5 min and then ethyl α -bromoisobutyrate (4.0 mg, 21 μ mol) was injected into the solution. The flask was stirred in an oil bath at 40 °C. When the conversion reached 95% as determined by ¹H-NMR spectrum, 1 ml of degassed DMF containing DEA (0.1 g, 0.54 mmol) was injected into the flask with stirring. The flask was then stirred at 40 °C overnight, and the solution was poured into methanol. The polymer was isolated and then further purified by dialysis against DMSO (MWCO: 3,500 Da). PDEA-PSN38 was obtained by lyophilization (0.28 g).

The molecular mass of PSN38 and PDEA-PSN38 was measured by GPC using DMF as the mobile phase. The PSN38 block had a molecular mass of 21 kDa with a PDI of 1.4, and the PDEA-PSN38 block copolymer had a molecular mass of 42 kDa with a PDI of 1.4.

PDEA-PSN38 (0.20 g) was dissolved in 1 ml of DMF and mixed with 1 ml DMF containing 0.30 g *meta*-chloroperoxybenzoic acid. The solution was stirred at room temperature for 1 h and then poured into ether. The isolated OPDEA-PSN38 (0.18 g) was washed with ether several times and dried under vacuum.

Synthesis of Cy5.5-OPDEA. DEA and 1 mol% 2-(*N*-Boc-amino)ethanol methacrylate were copolymerized and oxidized as described above to produce OPDEA containing 2-(*N*-Boc-amino)ethyl methacrylate units. The product (0.20 g) was then dissolved in 2 ml CH₂Cl₂ containing 0.5 ml TFA. After stirring at room temperature for 4 h, the solvent was removed by rotary evaporation. The obtained crude product was dissolved in deionized water and then dialysed against deionized water (MWCO: 3,500 Da). The OPDEA-NH₂ (82 mg) was obtained by lyophilization.

The obtained OPDEA-NH₂ was dissolved in 1 ml DMSO and then the Cy5.5-NHS solution (200 μ l, 4 mg ml⁻¹) was added. The solution was stirred at room temperature for 6 h in the dark, and then dialysed against methanol for 6 h and then against deionized water (MWCO: 3,500 Da) for 4 h. Cy5.5-OPDEA (37 mg) was further purified by a Sephadex G15 column to remove any residual fluorescent molecules and then lyophilized to obtain a light green powder. The Cy5.5 fluorescence intensity was measured by a microplate spectrophotometer (SpectraMax M2E, Molecular Devices) with 676 nm excitation and 694 nm emission. The Cy5.5 content of Cy5.5-OPDEA was determined to be 1.0 wt% calculated from the Cy5.5 fluorescence standard curve. Reacting OPDEA-NH₂ with RhoB isothiocyanate produced RhoB-labelled OPDEA (RhoB-OPDEA). Similarly, the Cy5.5-labelled PEG (Cy5.5-PEG) was obtained and purified as described above.

Labelling OPDEA-PSN38 and PEG-PSN38 with Cy5.5 or RhoB. The labelling of OPDEA-PSN38 and PEG-PSN38 were carried out by thiolysis of the SN38 phenol ester with thiol-functionalized Cy5.5 (Cy5.5-SH)⁴⁰. Cy5.5-NHS (0.2 mg), cysteamine hydrochloride (0.06 mg) and 20 μ l TEA were mixed in 1 ml anhydrous

DMSO. The solution was stirred at room temperature under a nitrogen atmosphere to obtain Cy5.5-SH solution. OPDEA-PSN38 (5.0 mg) was dissolved in 1 ml anhydrous DMSO and then mixed with 250 μ l of the above Cy5.5-SH solution. The mixture was stirred overnight at room temperature in the dark. The obtained OPDEA-Cy5.5-PSN38 solution was dialysed against methanol for 6 h and then against deionized water (MWCO: 3,500 Da). The OPDEA-Cy5.5-PSN38 (4.6 mg) was purified then obtained by lyophilization. The Cy5.5 content of OPDEA-Cy5.5-PSN38 was determined to be 1.0 wt% as mentioned above. PEG-PSN38 labelled with 1.0 wt% Cy5.5 (PEG-Cy5.5-PSN38) was prepared using the same procedures.

The labelling with RhoB to give purified OPDEA-RhoB-PSN38 was carried out similarly except for using RhoB-SH, which was prepared by the reaction of RhoB with cysteamine. The RhoB content was determined to be 1.0 wt% by measuring its fluorescence at the excitation of 555 nm.

Fabrication of OPDEA-PSN38 and PEG-PSN38 micelles. OPDEA-PSN38 (2 mg) dissolved in 0.5 ml DMF and deionized water (4.5 ml) was added dropwise to the solution with stirring. After 30 min, the solution was dialysed against deionized water (MWCO: 3,500 Da). The obtained solution was concentrated to the desired concentrations. The formed micelles of OPDEA-PSN38 had a diameter of 51 nm and ζ -potential of -10.5 mV at pH 7.4, as measured by dynamic light scattering. The OPDEA-Cy5.5-PSN38 and OPDEA-RhoB-PSN38 micelles were fabricated using the same procedures in the dark.

The PEG-PSN38 and PEG-Cy5.5-PSN38 micelles were fabricated using our previously reported method⁴⁰. The PEG-PSN38 diameter was 35 nm with a ζ -potential of -11.1 mV.

Synthesis of Cy5.5-PolyCP. Methanol (0.79 g, 24.7 mmol), Et₃N (2.50 g, 24.7 mmol), and acetonitrile (50 ml) were charged into a three-neck round-bottom flask under a nitrogen atmosphere. The mixture was cooled to -78°C and 2-chloro-2-oxo-1,3,2-dioxaphospholane (3.52 g, 24.7 mmol) in acetonitrile (20 ml) was added dropwise over 2 h. The reaction mixture was stirred at -78°C for 2 h and then was allowed to warm to room temperature for another 4 h. The reaction mixture was then cooled at -20°C for 2 h to precipitate the salts and then filtered. The filtrate was transferred into a Schlenk flask containing poly[(2-(*N,N*-dimethylamino)ethyl methacrylate)-*co*-(2-(*N*-Boc-amino)ethyl methacrylate)] (MW 10 kDa, 0.63 g) under nitrogen atmosphere. The resulting mixture was stirred overnight at 70°C , cooled to room temperature, concentrated, and then precipitated into an excess of ether. The crude product was then dissolved in 10 ml CH₂Cl₂ containing 2 ml TFA. After stirring at room temperature for 4 h, the mixture was concentrated by rotary evaporation and precipitated into an excess of ether. The obtained crude product was redissolved in deionized water and then dialysed against deionized water (MWCO: 3,500 Da). The polyCP-NH₂ (0.57 g, the amine-content of 1%) was obtained as a white powder by lyophilization.

PolyCP-NH₂ (0.1 g) was dissolved in 1 ml DMF, and then Cy5.5-NHS solution (250 μ l, 4 mg ml⁻¹) and Et₃N (20 μ l) were added. The solution was stirred overnight at room temperature in the dark and dialysed against methanol for 24 h and then against deionized water (MWCO: 3,500 Da) for 24 h. The Cy5.5-PolyCP was purified by a Sephadex G15 column and then lyophilized to obtain a light green powder.

Synthesis of Cy5.5-PolyPC. 2-Methacryloyloxyethyl phosphorylcholine (mPC, 0.30 g, 1.02 mmol) and 2-(*N*-Boc-amino)ethyl methacrylate (2.3 mg, 0.01 mmol) were copolymerized via ATRP as described above to produce poly[mPC-*co*-(2-(*N*-Boc-amino)ethyl methacrylate)] (MW: 20 kDa). The product (0.25 g) was then dissolved in 5 ml CH₂Cl₂ containing 1 ml TFA. After stirring at room temperature for 4 h, the mixture was concentrated by rotary evaporation and precipitated into an excess of ether. The obtained crude product was redissolved in deionized water and then dialysed against deionized water (MWCO: 3.5 kDa). Poly[mPC-*co*-(2-aminoethyl methacrylate)] (PolyPC-NH₂) (0.15 g, the amine-content of 1%) was obtained as a white powder by lyophilization.

The Cy5.5-PolyPC was synthesized and purified as described above.

Size and ζ -potential measurements. The materials (0.1 mg ml⁻¹) were dissolved in HEPES buffer (10 mM, pH 7.4) at designed pH for measurements. The sizes and ζ -potentials of materials were detected using Zetasizer Nano ZS (Malvern Instrument), equipped with a 4 mW He-Ne laser at a wavelength of 633 nm at 25°C .

Subcellular distribution of fluorescent-labelled materials observed by CLSM. The HepG2, A549, HeLa, HUVEC or ECDHCC-1 cells were seeded in a glass-bottom dish at a density of 1×10^5 cells per well and incubated for 24 h. According to the experimental needs, the organelles were stained as follows: the nuclei were stained with Hoechst 33342 (2 μ M) for 20 min, the lysosomes were stained with LysoTracker green (0.2 μ M) for 20 min, the endoplasmic reticulum was stained with ER tracker (0.2 μ M) for 20 min, Golgi apparatuses were stained with Golgi tracker (5 μ M) for 30 min, and the macropinosomes were stained with TRITC-Dex (7.0 μ M) for 4 h, and then incubated with fluorescent dye-labelled polymer, conjugate or micelles at designated concentrations. At designated timed intervals, cells were washed twice with PBS and imaged by CLSM. For time-lapse videos, cells were imaged immediately after adding fluorescent dye-labelled polymer/conjugate/micelles.

The excitation wavelengths were 405 nm for Hoechst 33342, 488 nm for LysoTracker green, TRITC-Dex, ER tracker or Golgi tracker, 543 nm for RhoB and 640 nm for Cy5.5. The imaging parameters were fixed in the same experiment. Pearson's correlation and the integrated fluorescence intensity were calculated by Image-Pro Plus software.

Observation of lysosomal lysis using acridine orange staining. HepG2 cells were plated onto glass-bottom dishes at a density of 1×10^5 cells per dish. The cells were pre-treated with 50 μ l PBS, PEG (50 μ g ml⁻¹), OPDEA (50 μ g ml⁻¹), OPDEA-SN38 (SN38-equivalent dose, 2.0 μ g ml⁻¹), OPDEA-PSN38 (SN38-equivalent dose, 2.0 μ g ml⁻¹) or chloroquine (0.6 μ mol l⁻¹) for 2 h. Then cells were incubated with acridine orange (0.6 μ mol l⁻¹) for 30 min, washed twice with PBS, and observed by CLSM with excitation at 543 nm. The imaging parameters were fixed for different groups.

Flow cytometry analysis of endocytosis inhibitor effects on cellular uptake. The cellular uptake of Cy5.5-labelled polymer/conjugate/micelles by HepG2, HUVEC, and ECDHCC-1 cells was analysed by flow cytometry: cells were grown in a 12-well plate at a density of 1×10^5 cells per well for 12 h, and then incubated with Cy5.5-labelled materials at designated concentrations. At timed intervals, cells were washed twice with PBS and digested by trypsin, and then centrifuged at 1,000 rpm for 3 min to remove the trypsin. The collected cells were resuspended in 0.5 ml PBS and analysed by flow cytometry. Every 10,000 cells were counted to determine Cy5.5-positive cells at the FL4 channel. The experiment was repeated three times independently; data were presented as mean $\pm$ s.d.; CellQuest Pro software was used for analysis.

For each inhibitor, filipin (7.5 μ M), wortmannin (5 μ M), chlorpromazine (30 μ M), cytochalasin D (5 μ M), monensin (50 μ M), brefeldin A (90 μ M) or nocodazole (30 μ M), cell were pre-treated for 1 h, and then Cy5.5-labelled materials were added for incubation. Temperature-effect experiments at 4°C were performed in a 4°C refrigerator. At timed intervals, the Cy5.5-positive cells were analysed by flow cytometry as mentioned above.

Inhibition of cellular uptake by heparin or low temperature observed by CLSM. HepG2 cells were seeded in a glass-bottom petri dish at a density of 8×10^4 cells per dish in 2 ml medium and incubated for 24 h. The medium was replaced with fresh medium containing 0.5 mg ml⁻¹ heparin. After 30 min incubation, 50 μ l Cy5.5-OPDEA was added and incubated at 37°C for 1 h (Cy5.5-equivalent dose, 0.5 μ g ml⁻¹). Then LysoTracker green (0.2 μ M) and Hoechst 33342 (2 μ M) were added and incubated for 30 min. The cells were washed twice with PBS and observed by CLSM. Similarly, the low-temperature effect was performed with precooled Cy5.5-OPDEA and incubated in a 4°C refrigerator.

HPLC analysis. Sample preparation: the whole blood, urine, major organs or tumour homogenate (50 μ l) was mixed with NaOH solution (0.1 M, 50 μ l) and incubated in a 60°C water bath for 10 h. The mixture was added with 400 μ l acetonitrile and centrifuged at 5,000 rpm for 5 min. The supernatant (50 μ l) was added with HCl solution (0.2 M, 50 μ l) and vortexed for 30 s. The mixture was centrifuged at 10,000 rpm for 5 min. The supernatant was collected for HPLC analysis.

Determination of the SN38 extraction efficiency: OPDEA-PSN38 (SN38-equivalent dose, 5 μ g) was separately added into the blood, urine, major organs or tumour homogenate (50 μ l) and mixed thoroughly. The samples were prepared and their SN38 concentrations were determined using HPLC as described above. The extraction efficiency of SN38 in the blood, urine, tumour or organs was calculated accordingly.

The HPLC system consisted of a 1525 binary HPLC pump, 2475 multi- λ fluorescence detector, and a 2998 photodiode array detector and a SunFire C18 column. The mobile phase of methanol/water (65/35) at a constant flow rate of 1.0 ml min⁻¹ was used. An aliquot of 20 μ l was injected for SN38 detection at 35°C .

Blood clearance. Female ICR mice were randomly grouped ($n=3$): (1) OPDEA-SN38, (2) PEG-SN38, (3) OPDEA-PSN38, and (4) PEG-PSN38. Each formulation (0.2 ml per injection) was injected intravenously at a 7.5 mg kg⁻¹ SN38-equivalent dose for (1) and (2) and a 10 mg kg⁻¹ SN38-equivalent dose for (3) and (4). At timed intervals, blood samples were collected from the orbit venous plexus and prepared for HPLC analysis as mentioned above.

In vivo determination of the total SN38 concentrations in tumour and organs. HepG2 tumour model was established by subcutaneously injecting 2×10^6 cells into 8-week-old female BALB/c-nude mice. After the mean volume of the tumours reached 70 mm³, the mice were randomly divided into two groups ($n=6$) and intravenous injected with the treatments. At 6 h and 24 h post-injection, mice were killed, and the tumours were dissected and weighed. The total SN38 contents in the tumour were analysed by HPLC as mentioned above.

The in vivo biodistribution and toxicity of OPDEA-PSN38 in mice. The in vivo biodistribution of the OPDEA-Cy5.5-PSN38 in tissues was analysed using the IVIS Spectrum system. Female BALB/c-nude mice bearing HepG2 tumours

of $\sim 100 \text{ mm}^3$ were randomly grouped ($n = 3$) and intravenously injected with OPDEA-PSN38 (SN38-equivalent dose, 20 mg kg^{-1}). At timed intervals, the mice were killed and their major organs, including heart, kidneys, spleen, lung, liver, and tumours were excised for ex vivo imaging. The Cy5.5 fluorescence intensity in each organ was calculated using NIS-Elements Microscope Imaging Software. Then, the dissected tissues were fixed with 4% paraformaldehyde and embedded in paraffin for H&E staining. The H&E-stained sections were observed by an optical microscope.

The in vivo serum and organ toxicology of polymers were tested by serum chemistry analysis. OPDEA-SN38, OPDEA-PSN38, CPT-11 or PBS was intravenously injected to female ICR mice (6–8 weeks old) at an SN38-equivalent dose of 20 mg kg^{-1} . The whole blood was collected from the orbit venous plexus 1 d and 7 d after injection, blood samples were tested at Zhejiang University Hospital. The data were presented as mean $\pm$ s.d. ($n = 3$).

Determination of the SN38 contents in nuclei of HepG2 tumour cells. HepG2 cells were seeded in a cell culture plate at a density of 1×10^7 in 15 ml medium and incubated for 24 h. Then cells were treated with OPDEA-PSN38 for 8 h at SN38-equivalent dose of $5 \mu\text{g ml}^{-1}$. The cells were washed twice with PBS, digested by trypsin and counted (2×10^7). The cells were divided into two parts, one was redistributed in 0.5 ml hypotonic solution (containing 10 mM NaCl, 2.5 mM MgCl₂ and 10 mM Tris-HCl) and disrupted by sonication for 5 min. Then, the mixture was added to 0.25 ml of 0.75 M sucrose buffer and centrifuged at 1,000g for 5 min to remove the unbroken cells. The supernatant was centrifuged at 1,000g for 15 min, and the precipitate (nuclei) was obtained. The other part was centrifuged at 1,000g for 5 min to collect cells. Finally, the total SN38 contents in nuclei and the cells were determined using HPLC as described above.

MTT assay. HepG2 cells were plated into 96-well plates (4,000 cells per well) and incubated for 24 h. The cells were incubated with different formulations for 48 h, then the medium in each well was replaced with 0.2 ml of fresh medium containing 0.5 mg ml^{-1} MTT for another 3 h incubation. The medium in each well was further replaced with 0.2 ml DMSO to dissolve the formazan crystals. The absorbance in each well was measured at 562 nm and 620 nm using a microplate spectrophotometer. Each drug concentration was tested in triplicate in three independent experiments. Cell viability was calculated from equation (1), where OD is the optical density of sample or control at the indicated wavelength:

$$\text{Cell viability (\%)} = \frac{\text{OD}_{\text{sample, 562 nm}} - \text{OD}_{\text{sample, 620 nm}}}{\text{OD}_{\text{control, 562 nm}} - \text{OD}_{\text{control, 620 nm}}} \times 100\% \quad (1)$$

Preparation of MTSs. MTSs were prepared using the hanging-drop method²³. HepG2 or HT29 cells were suspended in fresh DMEM medium (containing 0.12% w/v methylcellulose) at a density of 4×10^5 cells per ml. Cell suspensions (25 μl) were dropped on the lids of the cell culture plate to form uniform droplets, and 20 ml PBS was added to the plate to keep the droplets moist. The cells were incubated for 72 h and formed dense spheroids, which were transferred to agarose-coated (1% w/v in PBS) 96-well plate with one spheroid per well, and incubated for another 72 h to mature.

In vitro cytotoxicity assay using MTSs. The HepG2 MTSs were treated with OPDEA-PSN38 or PEG-PSN38 at different concentrations for 96 h. The resazurin was added to the plate at 0.1 mg ml^{-1} for an additional 6 h incubation. Every 100 μl of medium in each well was transferred to a black 96-well plate and the fluorescence intensity of the medium was measured with excitation at 530 nm using a microplate spectrophotometer. Each sample was tested in triplicate in three independent experiments. MTS viability was calculated from equation (2), where F is the fluorescence intensity of sample or medium, as indicated:

$$\text{MTS viability (\%)} = \frac{F_{\text{sample}} - F_{\text{blank medium}}}{F_{\text{control}} - F_{\text{blank medium}}} \times 100\% \quad (2)$$

To determine the effects of inhibitors of endocytosis and exocytosis on the cytotoxicity to the MTSs, HepG2 MTSs were pre-treated with PBS, wortmannin ($10 \mu\text{M}$) or brefeldin A ($10 \mu\text{M}$) for 2 h, then incubated with OPDEA-PSN38 for an additional 96 h. The MTS cell viability was measured according to the above procedures.

To visualize the morphological changes of MTS treated with OPDEA-PSN38 or PEG-PSN38, the HepG2 MTSs were treated with OPDEA-PSN38 or PEG-PSN38 at SN38-equivalent dose of $20 \mu\text{g ml}^{-1}$ for 96 h. The MTSs were freeze-sectioned into 10- μm -thick slices for H&E staining and observed by optical microscope. The integrated optical density of each slice was calculated by ImageJ software.

Observation of the MTSs by CLSM. MTSs were incubated with Cy5.5-OPDEA, Cy5.5-PEG, Cy5.5-PolyCR, or Cy5.5-PolyPC at Cy5.5-equivalent dose of $1.0 \mu\text{g ml}^{-1}$ for timed intervals. The MTSs were washed with PBS containing 0.5 mg ml^{-1} heparin and directly imaged using CLSM by Z-stack tomoscan at $20 \mu\text{m}$ intervals from the bottom to the middle of the spheroids. Furthermore, the MTSs were freeze-sectioned into 10- μm -thick slices, the nuclei were stained with Hoechst

33342 for CLSM imaging. The camera setting was fixed for different groups. The integration of Cy5.5 fluorescence density and linescan analysis was studied by ImageJ software.

Effects of inhibitors of endocytosis and exocytosis on the penetration of OPDEA in MTS: HepG2 and HT29 MTSs were separately treated with PBS, wortmannin ($2 \mu\text{M}$), Exo1 ($50 \mu\text{M}$), or brefeldin A ($10 \mu\text{M}$) for 2 h, and then incubated with RhoB-OPDEA (RhoB-equivalent dose, $5 \mu\text{g ml}^{-1}$) for 4 h. The MTSs were imaged and analysed as described above.

In vivo real-time imaging of Cy5.5-OPDEA extravasation from blood vessels and tumour penetration. The procedure followed a reported method⁷⁴ with modifications. HepG2 cells (1×10^5) were inoculated subcutaneously near the subcutaneous venous vessel in the abdomen of female BALB/c-nude mice (6–8 weeks old). When the tumour reached around 6 mm^3 , the mice were anaesthetized and the tumour was exposed and fixed on a microscope slide using Histoacryl. After intravenous injection of 200 μl of Cy5.5-OPDEA or Cy5.5-PEG (Cy5.5-equivalent dose, 0.5 mg kg^{-1}), the tumour was imaged at timed intervals with fixed settings using CLSM. At 2 h post-injection, the free Cy5.5-OPDEA or Cy5.5-PEG in the blood vessel was washed out by heart perfusion, then the tumour was frozen-sectioned into 10- μm -thick slices. Slices were treated with a drop of Hoechst 33342 for 30 min to stain nuclei, then washed twice with TBST and observed by CLSM to image the distribution of Cy5.5-OPDEA or Cy5.5-PEG in tumours.

The extravasation of Cy5.5-OPDEA in a subcutaneous 4T1 tumour and a HepG2 tumour inoculated on one of the ears were evaluated using a similar procedure. The ear was fixed on a glass slide and imaged directly.

The extravasation of Cy5.5-OPDEA from normal blood vessels was observed via the subcutaneous venous vessel in the abdomen of a healthy mouse.

Ex vivo analysis of tumour penetration of OPDEA. Mice bearing the HepG2 tumours of about 10 mm^3 in size were injected intravenously with 200 μl Cy5.5-OPDEA or Cy5.5-PEG at a Cy5.5-equivalent dose of 0.5 mg kg^{-1} . At 24 h post-injection, the mice were killed and the tumours were frozen-sectioned into 10- μm -thick slices. For each tumour, three adjacent slices were selected and treated as follows. One slice was directly observed by CLSM to image the distribution of Cy5.5-OPDEA or Cy5.5-PEG. The second slice was H&E stained and observed by an optical microscope to identify the necrotic regions. The third slice was observed by CLSM to determine the distribution of blood vessels after immunohistochemical staining with CD31 as follows. The slice was first treated with the block solution (0.5% goat serum, 10% nonfat milk in antibody-carrier solution) for 2 h at room temperature, then washed twice with PBS solution containing 0.3% Tween-20. Rat-anti-mouse CD31 antibody at 1:50 dilution was dropped onto the slice and incubated at 4°C for 12 h. The section was washed twice with PBS solution containing 0.3% Tween-20, followed by incubation with Dylight-405 conjugated goat-anti-rat IgG (1:50 dilution) at 4°C for another 12 h. Finally, the slice was washed twice with PBS solution containing 0.3% Tween-20. The distribution of Cy5.5-OPDEA or Cy5.5-PEG was imaged by CLSM and analysed using ImageJ software.

Photoacoustic imaging of the intratumoral distribution of OPDEA-Cy5.5-PSN38. Female BALB/c-nude mice bearing subcutaneous HepG2 tumours of $\sim 10 \text{ mm}^3$ were established as described above. The mice were intravenously injected with 200 μl of OPDEA-Cy5.5-PSN38 at Cy5.5-equivalent dose of 0.5 mg kg^{-1} . At 6 h post-injection, the mice were anaesthetized and the tumour was scanned repeatedly at excitation wavelengths of 680, 700, 756, 804 and 924 nm using a photoacoustic imaging system (Vevo LAZR LZ-400). The distributions of OPDEA-Cy5.5-PSN38 and oxygenated haemoglobin were imaged, the data were analysed using photoacoustic imaging software.

Distribution of OPDEA-RhoB-PSN38 micelles in tumour hypoxic areas. Female BALB/c-nude mice bearing subcutaneous HepG2 tumour of about 10 mm^3 were injected intravenously with 100 μl of HypoxiSense 680 (20 nmol ml^{-1} , NEV11070, Perkin Elmer). At 18 h post-injection, mice were injected intravenously with 200 μl of OPDEA-RhoB-PSN38, RhoB-equivalent dose of 0.5 mg kg^{-1} . After another 6 h, the mice were killed, and their tumours were excised and frozen sectioned into 10- μm -thick slices. The fluorescence was observed by CLSM with excitation at 640 nm for HypoxiSense 680 and 543 nm for OPDEA-RhoB-PSN38.

For comparison with PEG-Cy5.5-PSN38, mice were injected intravenously with 200 μl PEG-Cy5.5-PSN38 at a Cy5.5-equivalent dose of 0.5 mg kg^{-1} . After 4 h, 100 μl of pimonidazole (60 mg kg^{-1}) was injected intravenously. After another 2 h, the mouse was killed. The tumour was dissected, frozen-sectioned into 10- μm -thick slices and fixed in cold acetone for 10 min. The HypoxyProbe-1 (1:100 dilutions) was dropped onto the slices followed by 12 h incubation at 4°C . The slices were washed twice with TBST and stained with Hoechst 33342. The fluorescence was observed by CLSM with excitation at 488 for HypoxyProbe-1 and 647 nm for PEG-Cy5.5-PSN38.

RBC haemolysis experiment. Whole blood was collected into EDTA-pre-treated tubes from the orbit venous plexus from BALB/c-nude mice (female, 6–8 weeks old), and washed three times with erythrocyte buffer (containing 150 mM NaCl and 6 mM glucose in deionized water) by centrifugation (300g for 5 min). The

obtained RBCs were diluted to 5×10^8 RBCs per ml with erythrocyte buffer at the designated pH. OPDEA, OPDEA-SN38, OPDEA-PSN38 or PEG was dissolved in erythrocyte buffer and added to the RBC suspensions at desired concentrations (10 or $50 \mu\text{g ml}^{-1}$). RBCs treated with Triton X-100 (0.1 mg ml^{-1}) was used as a 100% haemolysis control. All the treated samples were gently vortexed, incubated at 37°C for 1 h, and then centrifuged at $2,500 \text{ rpm}$ for 5 min. The absorbance of the supernatants was measured at 541 nm using a microplate spectrophotometer. The haemolysis ratio was calculated from equation (3):

$$\text{Haemolysis ratio} = \frac{\text{OD}_{\text{sample}} - \text{OD}_{0\% \text{ control}}}{\text{OD}_{100\% \text{ control}} - \text{OD}_{0\% \text{ control}}} \quad (3)$$

In vitro adsorption of Cy5.5-labelled polymers on RBCs. Cy5.5-OPDEA, Cy5.5-PolyCP or Cy5.5-PolyPC solution ($50 \mu\text{l}$, Cy5.5-equivalent dose, $0.5 \mu\text{g ml}^{-1}$) was separately added to 1 ml of the RBC suspension (5×10^8). After incubation at 37°C in the dark for 3 h, each sample was divided into two parts. One aliquot was centrifuged to isolate the RBCs, and the other was manually washed once or twice with erythrocyte buffer by extrusion (20 times per washing) using a syringe (needle diameter: 0.7 mm). The RBCs were collected by centrifugation at $300g$ for 2 min. Fluorescent images of the RBCs were taken using CLSM with excitation at 640 nm .

In vitro desorption kinetics of OPDEA-Cy5.5-PSN38 from RBCs. RBCs suspension (0.5 ml , 5×10^8 RBCs per ml) was incubated with OPDEA-Cy5.5-PSN38 for 1 h at Cy5.5-equivalent dose of $0.25 \mu\text{g ml}^{-1}$, then gently washed twice with erythrocyte buffer to remove the free OPDEA-Cy5.5-PSN38. The resulted RBCs binding with OPDEA-Cy5.5-PSN38 (OPDEA-Cy5.5-PSN38@RBC) were resuspended in 0.5 ml erythrocyte buffer and incubated in the dark for 0, 5, 30 or 60 min. The RBCs were isolated by centrifugation ($2,500 \text{ rpm}$ for 5 min) and analysed by flow cytometry.

Colocalization of Cy5.5-OPDEA or Cy5.5-PEG with RBC membranes. The colocalization of OPDEA and RBC membranes was analysed following the described procedures⁷⁵. The RBC suspension (0.5 ml , 5×10^8 cells per ml) was stained with FITC-WGA ($0.1 \mu\text{M}$) for 5 min in the dark. Cy5.5-OPDEA or Cy5.5-PEG solution ($50 \mu\text{l}$, Cy5.5-equivalent dose, $0.5 \mu\text{g ml}^{-1}$) was separately added to FITC-WGA-labelled RBCs suspension and incubated for 1 h in the dark. The sample was divided into two parts: one aliquot was washed three times with erythrocyte buffer by extrusion using a syringe and the other aliquot was used directly. Fluorescent images were taken using CLSM with excitation at 488 nm for FITC-WGA and 640 nm for Cy5.5-OPDEA or Cy5.5-PEG.

In vivo distribution of Cy5.5-OPDEA in plasma, RBCs and urine. Healthy female ICR mice (8 weeks old) were injected intravenously with $300 \mu\text{l}$ Cy5.5-OPDEA or OPDEA-Cy5.5-PSN38 at a Cy5.5-equivalent dose of 0.5 mg kg^{-1} ($n = 3$). At timed intervals, a drop of urine was collected by gently massaging the bladder, and a drop of blood was collected from the orbit venous plexus. The blood samples were centrifuged ($2,500 \text{ rpm}$, 5 min) to separate the plasma and RBCs, and the collected RBCs were gently washed once and resuspended in erythrocyte buffer. The samples were distributed to a black 96-well-plate to detect the Cy5.5 fluorescence intensity of plasma, RBCs and urine using the IVIS Spectrum imaging system, and the corresponding Cy5.5-OPDEA or OPDEA-Cy5.5-PSN38 concentrations were quantified according to standard curves. In addition, fluorescent images of the above RBCs at 2 min post-injection were observed by CLSM with excitation at 640 nm .

Determination of total SN38 excreted in urine. Female ICR mice (6–8 weeks old) were injected intravenously with $300 \mu\text{l}$ PBS or OPDEA-PSN38 at SN38-equivalent dose of 10 mg kg^{-1} ($n = 3$). The mice were individually housed in metabolic cages, fed with chow and water ad libitum. The 0–6 h and 6–24 h urine were collected for analysis. The SN38 content in the urine sample was analysed using HPLC as mentioned above.

Adsorption of OPDEA on a PC-like-molecule-coated surface. 2-Methacryloyl oxyethyl phosphorylcholine (7.2 mg) was dissolved in 2 ml DMSO. 1,2-Ethanedithiol ($31 \mu\text{l}$, 0.5 g ml^{-1} in DMSO) was added to the solution and stirred in the dark at room temperature for 4 h. The solution was dropped on half the side of the gold-coated glass slides and incubated for 1 h. The slides were then gently washed with PBS and dried. The slides were then treated with Cy5.5-OPDEA or Cy5.5-PEG (Cy5.5-equivalent dose, $2 \mu\text{g ml}^{-1}$) for 3 min, and gently washed with PBS and observed by CLSM.

SPR experiment. The high-resolution SPR experiment was studied followed by a previous literature⁷⁶. The gold (Au)-coated slides were immersed in $1.0 \times 10^{-5} \text{ mol l}^{-1}$ 11-mercaptopundecanoic acid (MUA) ethanol solution, then successively washed with ethanol and deionized water. After drying under nitrogen flow, MUA-Au-coated slides were obtained.

The MUA-Au-coated slides were immersed into PC or PE solution ($1 \times 10^{-3} \text{ mol l}^{-1}$ in PBS) for 1 h and washed with PBS and with water, then dried under nitrogen flow. The PC-MUA-Au or PE-MUA-Au coated slides were set on the SPR instrument. The OPDEA or PEG solution ($0.5 \mu\text{g ml}^{-1}$) was injected into

the SPR instrument at a flow rate of 0.1 ml h^{-1} . The shift of the absorption peak ($\Delta\theta$) signal was recorded for more than 600 s.

Preparation of RBC, HUVEC, ECDHCC-1 and HepG2 cell membranes. RBC membranes were prepared as follows: RBCs were incubated with RBC lysis buffer for 15 min at 4°C . Haemoglobin was removed by centrifugation ($3,500 \text{ rpm}$ for 30 min), and the erythrocyte ghosts were gently washed three times with erythrocyte buffer and suspended in HEPES (5 mM , pH 7.4). The erythrocyte ghosts were sonicated for 20 min and extruded through 200 nm polycarbonate porous membranes (Avanti mini extruder, AvantiPolar Lipids). The BCA Protein Assay Kit was used to measure the concentration of the membrane proteins to calculate the lipid concentration (RBCs ghost membrane comprises about 20% protein and 80% lipid⁷⁷).

The HUVEC, ECDHCC-1 and HepG2 cell membrane solutions were prepared as follows: cells were cultured to exponential phase, washed twice with PBS, digested with trypsin, centrifuged to remove the trypsin, and then redistributed in 5 ml deionized water and disrupted by sonication for 5 min. The homogenate was centrifuged at $1,000g$ for 5 min and the supernatant was isolated followed by centrifugation at $10,000g$ for 30 min. The supernatant was again isolated and centrifuged at $100,000g$ for 30 min. The precipitate (cell membrane) was obtained and resuspended in HEPES buffer. The lipid concentration was calculated as mentioned above.

ITC experiments. ITC experiments were performed at 37°C and atmospheric pressure. Twenty-nine sequential injections were made for each ITC experiment. The volume of the first injection was $4 \mu\text{l}$, and then a constant volume of OPDEA (100 mM , $10 \mu\text{l}$ per injection) was injected into a cell filled with PE or PC solution (10 mM in ethanol:water 4:1 v/v). Each titration produced a reaction heat profile. The dilution heat was subtracted from the heat determined in the titration experiments to obtain the net reaction heat. The results were analysed using the 'one set of binding sites' model to give the stoichiometry (n), association constant (K_a), standard molar reaction enthalpy (ΔH°) and entropy changes (ΔS°).

The ITC assay with the RBC membrane was performed in HEPES buffer (5 mM , pH 7.4) at 37°C and atmospheric pressure. Fifty sequential injections were made for each ITC experiment. The first injection was $2 \mu\text{l}$, then a constant volume ($5 \mu\text{l}$ per injection) of OPDEA (25 mM in HEPES) or PEG (25 mM in HEPES) was injected into the cell filled with RBC membrane solution (lipids equivalent dose, 2 mmol l^{-1}).

Similarly, the ITC assay with the HUVEC, ECDHCC-1 or HepG2 cell membranes was performed in HEPES buffer (5 mM , pH 7.4) at 37°C and atmospheric pressure. Twenty-five sequential injections were made for each ITC experiment. The first injection was $0.5 \mu\text{l}$, then a constant volume ($2.0 \mu\text{l}$ per injection) of OPDEA (25 mM in HEPES) was injected into the cell filled with cell membrane solution (lipids equivalent dose, 1 mmol l^{-1}).

Distribution of OPDEA-PSN38 to RBCs and ECs. The cellular uptake of Cy5.5-labelled materials in the presence of both RBCs and endothelial cells (HUVECs or ECDHCC-1 cells) were cultured in a 12-well plate at a density of 1×10^5 cells per well. The RBC suspension (0.5 ml , $5 \times 10^8 \text{ ml}^{-1}$) was added into the 12-well plate and incubated with HUVECs or ECDHCC-1 cells. OPDEA-Cy5.5-PSN38 or PEG-Cy5.5-PSN38 at a Cy5.5-equivalent dose of $0.25 \mu\text{g ml}^{-1}$ was added to each well and incubated for 5, 30 or 60 min. RBCs were isolated from the medium by centrifugation, and ECDHCC-1 was collected by digestion. The collected Cy5.5-positive RBCs and ECs were analysed by flow cytometry.

In vivo translocation of OPDEA from RBCs for tumour extravasation. The RBC suspension (5×10^{10} in 0.2 ml erythrocyte buffer) was incubated with Cy5.5-OPDEA (Cy5.5-equivalent dose, 0.1 mg ml^{-1}) at 37°C in the dark for 3 h, and gently washed twice with erythrocyte buffer to remove the free Cy5.5-OPDEA. The resulting Cy5.5-OPDEA@RBC was resuspended in 0.2 ml erythrocyte buffer and injected intravenously into the HepG2 tumour-bearing mice ($\sim 6 \text{ mm}^3$). The Cy5.5-OPDEA extravasation from blood vessels into the tumour tissues was visualized by real-time imaging with CLSM.

Exo1 effect on Cy5.5-OPDEA penetration in tumours: $10 \mu\text{l}$ Exo1 ($150 \mu\text{M}$) was injected into tumours ($\sim 50 \text{ mm}^3$) of HepG2 tumour-bearing mice. After 12 h, $200 \mu\text{l}$ of Cy5.5-OPDEA@RBC was injected intravenously followed by real-time imaging as described above.

Trans-endothelial transport measurements. HUVECs or ECDHCC-1 cells were seeded onto transwell polycarbonate cell culture inserts (24-well, Corning) at a density of 5×10^5 cells per well. During long-term culture, the media in both apical and basolateral compartments were replaced every day, and the trans-endothelial electrical resistance was measured by epithelial volt/ohm meter (Millicell ERS-2, Millipore) to monitor the integrity of the cell monolayer. After the trans-endothelial electrical resistance value was above $30 \Omega\text{-cm}^2$, the medium in the apical compartment was replaced with 0.2 ml fresh medium containing polymer (Cy5.5-OPDEA or Cy5.5-PEG, Cy5.5-equivalent dose, $0.8 \mu\text{g ml}^{-1}$), micelle (OPDEA-Cy5.5-PSN38 or PEG-Cy5.5-PSN38, Cy5.5-equivalent dose, $1.0 \mu\text{g ml}^{-1}$) or OPDEA-Cy5.5-PSN38@RBC (prepared as mentioned above), and the medium

in the basolateral compartment was replaced with 0.5 ml fresh medium. The cells were incubated under static or flow conditions (approximately 0.6 or 1.8 ml min⁻¹). At timed intervals, 200 µl medium was sampled from the basolateral compartment, and the same volume of fresh medium was supplemented. The polymer concentration in the sampled solution was measured with a microplate spectrophotometer. RBCs were isolated from the sampled medium by centrifugation, and ECDHCC-1 was collected by digestion. The collected Cy5.5-positive RBCs and endothelial cells were analysed by flow cytometry.

To investigate the effect of inhibitors on the trans-endothelial transport, the endo- and exocytosis inhibitors cytochalasin D (5 µM), monensin (50 µM), brefeldin A (90 µM) or nocodazole (30 µM) were separately added to the medium both in apical and basolateral compartments during the experiment. Experiments at 4 °C were performed in a 4 °C refrigerator. The polymer concentration in the sampled solution was measured using a microplate spectrophotometer as mentioned above.

Transmission electron microscopy. In vitro HUVEC or ECDHCC-1 cells: cells were seeded in a 12-well plate at a density of 1×10^5 cells per well. The cells were treated with OPDEA-Cy5.5-PSN38 or PEG-Cy5.5-PSN38 (Cy5.5-equivalent dose, 0.4 µg ml⁻¹) for 4 h, and the medium was replaced with fresh medium for another 4 h incubation. The cells were washed twice with PBS and digested by trypsin. The obtained cells were fixed with 2.5% glutaraldehyde at 4 °C for 12 h, washed with PBS (0.1 M, pH 7.0), embedded in 1% agarose B, and trimmed into small sections (about $3 \times 3 \times 3$ mm³). The samples were then treated with 1% osmium tetroxide (in 100 mM cacodylate buffer) for 1 h, dehydrated with ethanol and acetone, and embedded in Spurr resin. The ultrathin sections (70–90 nm) were prepared using an ultramicrotome (UC7, Leica), then stained with lead citrate and uranyl acetate for observation (JEM-1230, JEOL).

In vivo tumour blood capillary ECs: nude mice bearing subcutaneous HepG2 tumours of ~150 mm³ were randomly divided into two groups ($n = 3$). OPDEA-PSN38 or PEG-PSN38 (200 µl, SN38-equivalent dose, 50 mg kg⁻¹) was injected through the tail vein. After 6 h, the mice were anaesthetized and perfused with fixative (containing 4% PVP, 3% glutaraldehyde, 1% sucrose and 0.05% calcium chloride in 75 mmol l⁻¹ sodium cacodylate buffer, pH 7.1) by cardiac injection to fix the tumours. The tumour vasculature was collected and fixed overnight at 4 °C. The samples were trimmed into small sections (about $1 \times 3 \times 1$ mm³) and then treated with 1% osmium tetroxide (diluted in 100 mM cacodylate buffer) for 2 h, dehydrated with ethanol and acetone, and embedded in Spurr resin. Ten-micrometre-thick sections were prepared (LKB 11800 Pyramitome) and stained with toluidine blue to select the desired region. The ultrathin sections (50–100 nm) were prepared using an ultramicrotome (UC7, Leica) then stained with lead citrate and uranyl acetate for observation (H-7650, Hitachi).

Determination of intercellular transport of RhoB-OPDEA. The transcytosis of OPDEA in tumour cells was studied using the previously reported method^{48,78}. HepG2 or HT29 cells were seeded at a density of 2×10^5 cells per well in the glass-bottom Petri dish and incubated for 24 h. The first-batch cells were cultured with RhoB-OPDEA at a RhoB-equivalent dose of 5 µg ml⁻¹ for 5 h, and washed twice with PBS before imaging. These cells were cultured in a fresh medium for 12 h, then washed and imaged; the culture medium was collected and used to culture the second batch of new cells for 12 h, and the cells were imaged. Similarly, the second-batch cells were then cultured, and the culture medium was further used to culture the third-batch fresh cells for 12 h before imaging. All the cells were imaged by CLSM with excitation at 543 nm for RhoB with fixed settings.

Establishment of tumour models. Establishment of subcutaneous cell-derived xenograft (CDX) models: HepG2 or HT29 cells (1×10^7) were injected subcutaneously into the armpit of 6-week-old female nude mice. When the tumours reached the desired volumes, the mice were randomly grouped and treatments were initiated. The tumour volumes (V) were calculated as $V = 0.5ab^2$, where a is the longest length and b is the longest width of the tumour.

Establishment of hepatocellular carcinoma orthotopic model: HepG2-Luci cancer cells (4×10^6) were injected subcutaneously into the armpit of nude mice (female, 6-week-old). When the tumour volumes reached ~100 mm³, the tumours were collected and resected under aseptic conditions. Necrotic tissues were removed, and viable tissues were cut and minced into ~1 mm³ pieces. A transverse incision was made below the right costal margin of the anaesthetized mice, and the tumour fragment was gently transplanted into the incision within the liver parenchyma. The abdominal wound was sutured with absorbable surgical sutures. On the seventh day after orthotopic tumour implantation, the mice were divided into three groups with similar bioluminescence intensity.

Establishment of PDX models: surgically resected human liver cancer tissues were obtained from patients with cancer following the protocol approved by the First Affiliated Hospital School of Medicine of Zhejiang University. The tissue generation resected from the patient is termed F0, with subsequent generations numbered consecutively. Fresh tumour tissue (F0) was cut into ~2 mm³ fragments and implanted into the flanks of NOD/SCID mice (4 week old, male). When the tumour volumes reached ~200 mm³ (F1), the mice were killed, and tumour

fragments of ~2 mm³ were implanted into the right flanks of athymic BALB/c-nude mice (4 week old, male). When the tumour volumes reached ~200 mm³ (F2), the mice were killed and tumour fragments (~2 mm³) were implanted into the flanks of BALB/c-nude mice (4 week old, male); the tumours that formed were denoted as F3. Generally, F2–F4 PDX tumours can be expanded for drug tests because they maintain the original tumour architecture and histological characteristics⁷⁹.

In vivo antitumour studies. Antitumour activity of free OPDEA polymer against HT29 or HepG2 PDX: nude mice bearing ~70 mm³ HepG2 tumours were randomly grouped; treatments were initiated by intravenous injection of PBS or OPDEA (200 mg kg⁻¹) every 3 d for 3 treatments ($n = 5$ for HT29 and $n = 4$ for HepG2). The mice were weighed and the tumour volumes were measured every 3 d.

Antitumour activity of OPDEA-SN38 or PEG-SN38 conjugates against HepG2 CDX tumours: nude mice bearing HepG2 tumours of ~80 mm³ ($n = 6$) were injected intravenously with PBS, OPDEA-SN38 or PEG-SN38 conjugates (SN38-equivalent dose, 7.5 mg kg⁻¹) every 3 d for 6 treatments. The tumour growth curves were calculated and the mice were weighed every 3 d.

Antitumour activity of OPDEA-PSN38 micelles: nude mice bearing HepG2 CDX tumours of ~70 mm³ were randomly grouped ($n = 6$) and injected intravenously with (1) PBS, (2) OPDEA-PSN38 or (3) PEG-PSN38 at (SN38-equivalent dose, 10 mg kg⁻¹) every 3 d, 5 times. The mice were weighed and the tumour volumes were measured every 3 d.

The antitumour activity of OPDEA-PSN38 against HT29 CDX tumours was compared with liposomal encapsulated CPT-11 (LipoCPT-11) as follows. Nude mice bearing HT29 tumours of ~100 mm³ were randomly grouped ($n = 5$) and the treatments were initiated by intravenous injection of PBS, LipoCPT-11 or OPDEA-PSN38 (SN38-equivalent dose, 20 mg kg⁻¹) every 3 d for 5 treatments. The mice were weighed and the tumour volumes were measured every 3 d.

Antitumour activity of OPDEA-PSN38 against large CDX tumours: Nude mice bearing HepG2 or HT29 tumours of ~500 mm³ were randomly grouped; treatments were initiated by intravenous injection with (1) PBS, (2) CPT-11, and (3) OPDEA-PSN38 at an SN38-equivalent dose of 20 mg kg⁻¹ every 3 d twice in HepG2 CDX tumours ($n = 4$), and for 6 treatments in HT29 CDX tumours ($n = 6$). The mice were weighed and the tumour volumes were measured every 3 d.

The antitumour activity of OPDEA-PSN38 against HepG2-Luci orthotopic tumour model: mice bearing orthotopic tumours were randomly divided into three groups ($n = 3$) and treated with (1) PBS, (2) OPDEA-PSN38, or (3) CPT-11 at an SN38-equivalent dose of 20 mg kg⁻¹ every other day for 4 treatments. The mice were intraperitoneally injected with 100 mg kg⁻¹ D-luciferin to monitor the in vivo bioluminescence of tumour using the IVIS Spectrum system every 5 d.

The antitumour activity of OPDEA-PSN38 against PDX models: when the tumours reached ~100 mm³, the PDX tumour-bearing mice were randomly grouped ($n = 5$) and treated with (1) PBS, (2) CPT-11 (SN38-equivalent dose, 10 mg kg⁻¹), (3) CPT-11 (SN38-equivalent dose, 20 mg kg⁻¹), or (4) OPDEA-PSN38 (SN38-equivalent dose, 10 mg kg⁻¹) every other day 3 times. The mice were weighed and the tumour volumes were measured every other day.

Statistical analyses. All data were presented as mean $\pm$ s.d. from $n \geq 3$ independent biological replicates unless otherwise stated. Graphing and statistical analyses were performed with GraphPad Prism version 8, Origin Pro version 9.1 and/or Microsoft Office Excel 2019. Statistical significance was determined between pairs of data with an unpaired two-tailed Student's t -test for continuous variables and one-way ANOVA with Tukey's correction and 95% confidence intervals for categorical variables. The specifics of the statistical tests and number of replicates are stated in the figure legends. The threshold for significance in all tests was $P < 0.05$. For principal component correlation analysis, Pearson's correlation coefficient (R) was used to assess the significance of the correlation.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The main data supporting the results in this study are available within the paper and its Supplementary Information. The raw and analysed datasets generated during the study are too large to be publicly shared, but are available for research purposes from the corresponding author on reasonable request.

Received: 24 April 2020; Accepted: 16 February 2021;
Published online: 15 April 2021

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Acknowledgements

This work received support from the National Natural Science Foundation (51833008) of China and Zhejiang Key Research Program (2020C01123). We thank H. Cui of Johns Hopkins University for his generous instructions and help, and H. Xu for the design of the scheme in Fig. 1. We acknowledge support from the Bio-ultrastructure Analysis Laboratory of the Analysis Center of Agrobiology and Environmental Science of Zhejiang University in sample preparation and transmission electron microscopy.

Author contributions

Y.S. conceived and supervised the project. S.C., Y.Z., W.F., Q.Z. and Y.P. performed the biological experiments and analysed the data; J.X., Jinqiang Wang, Y.G. and Z. Zhang performed the synthesis; G.W. did the transmission electron microscopy imaging and H&E staining; R.S. did the ITC assay; J.L. performed calculations; Jianguo Wang, J.Z. and X.X. established the PDX models; D.Y. and X.Y. performed the SPR experiment; H.C. and Z.L. helped with discussion of the synthesis and mechanism; H.J., J.T. and Z. Zhou helped with supervision of experiments and writing. All authors discussed the results and participated in writing the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41551-021-00701-4>.

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Peer review information *Nature Biomedical Engineering* thanks Daryl Drummond and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

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Software and code

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Data collection

Flow Cytometry: BD CellQuest™ Pro software (version 5.1)
NMR: MestReNova software (version 11.0.4)
Size/Zeta potential: Zetasizer software (version 7.02)
CLSM images: NIS Elements Viewer software (version 4.11)
Fluorescence intensity: SoftMax Pro (version 5.4)
In vivo bioluminescence/fluorescence images: Living image software (version 4.0)
ITC: Origin software (version 9.1)

Data analysis

All statistical analyses were analysed by GraphPad Prism (version 8.3.0), Microsoft office Excel (version 2019) or Origin Pro (version 9.1).
Flow cytometry data were analysed by FlowJo_V10 software (version 10.7.1).
Fluorescence images were analysed by ImageJ (version 1.52) and Image-Pro Plus (version 6.0).

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No sample-size calculations were performed, and sample sizes were chosen on the basis of the throughput of the technique used. Sample sizes were sufficient to show the same trends between the three or more replicates performed for each experiment. Owing to the limited availability of the photoacoustic imaging system, a single measurement was carried out for photoacoustic imaging of tumor-bearing mice. In particular, the in vivo efficacy studies were performed with 3–6 mice per group predicted by G-Power and/or consistent with previously published works, per standard protocols and to minimize the number of animals used.
Data exclusions	No data were excluded.
Replication	All attempts at replication were successful. Findings were replicated as indicated in the figure legends.
Randomization	Samples and cells were chosen at random for each experimental group. Mice were assigned to the treated or untreated groups by litter. No covariates were controlled for.
Blinding	Polymer/conjugate/micelles were prepared by J.X., Y.G., Z.Z. and J.Q.W. Given the need to properly synthesize the materials, the researchers were not blinded to the components that were used to synthesize each formulation. In particular, for TEM imaging, patient-derived xenograft treatments, surface plasmon resonance experiment and photoacoustic imaging analysis, the collaborators (G.W., R.S., J.G.W., J.Z., X.X., D.Y. and X.Y.) were blinded to the group allocation. For other experiments, the investigators were aware of the group allocation. Blinding was not necessary because no placebo group was present and the readout (data collected by instruments such as NMR, Zetasizer, flow cytometer, microplate reader, IVIS Viewer and HPLC etc.) can not be influenced by a biased researcher.

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Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

1. Purified Rat Anti-Mouse CD31: BD Pharmingen; Cat#. 550274; Lot#. 51627341; Monoclonal (MEC 13.3); Dilution 1:50.
2. DyLight 405AffiniPure Goat Anti-Rat IgG (H+L): Jackson Immuno Research Laboratories; Cat#112-475-003. RRID AB#2338303;

Polyclonal; Dilution: 1:50.

3. 4.3.11.3 mouse FITC-Mab: Hypoxiprobe, Inc.; Cat# HP6-100kit; Polyclonal (clone 4.3.11.3); Dilution 1:50.

Validation

We have only used antibodies from commercial sources. The suppliers have a standard validation system to ensure reproducibility, and validation data are available online at the manufacturers' websites.

1. Rat-anti-mouse CD31

<https://www.bdbiosciences.com/cn/applications/research/stem-cell-research/cancer-research/mouse/purified-rat-anti-mouse-cd31-mec-133/p/550274>

2. DyLight 405 AffiniPure Goat Anti-Rat IgG (H+L)

<https://www.jacksonimmuno.com/catalog/products/112-475-003/Goat-Rat-IgG-HL-DyLight-405>

3. 4.3.11.3 mouse FITC-Mab

<http://site.hypoxyprobe.com/HP-Green.pdf>

Eukaryotic cell lines

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Cell line source(s)

HepG2 (human hepatocellular carcinoma), A549 (human lung carcinoma), HeLa (human cervix adenocarcinoma), HT29 (human colorectal adenocarcinoma), HUVEC (human umbilical vein endothelial cell) and 4T1-GFP (mouse mammary carcinoma expressing green fluorescent protein) cell lines were obtained from the Cell Bank of the Chinese Academy of Sciences (Shanghai, China). ECDHCC-1 cell (hepatocellular carcinoma derived endothelial cell) was purchased from Fusheng Industrial Co., Ltd. (Shanghai, China).

Authentication

All cell lines used in this study were authenticated by the Cell Bank of the Chinese Academy of Science and Fusheng Industrial Co., Ltd. (Shanghai, China).

Mycoplasma contamination

The cell lines were regularly tested for mycoplasma contamination, and no mycoplasma contamination was found.

Commonly misidentified lines (See [ICLAC](#) register)

None of the cell lines used are listed in the ICLAC list.

Animals and other organisms

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Laboratory animals

BALB/c-nude mice (female, 4–8 weeks old, 18 g) and ICR mice (female, 4–6 weeks old, 18 g) were purchased from Shanghai Slaccas Laboratory Animal Co., Ltd. (Shanghai, China). NOD/SCID mice (male, 4 weeks old, 18 g) were purchased from Model Animal Research Center of Nanjing University (Nanjing, China).

Wild animals

The study did not involve wild animals.

Field-collected samples

The study did not involve samples collected from the field.

Ethics oversight

All animal experiments were carried out according to the protocols approved by the Zhejiang Academy of Medical Sciences and UKCCCR guidelines. Surgically resected human liver cancer tissues were obtained from liver cancer patients #P25 and #P51 following the protocol approved by the First Affiliated Hospital of Zhejiang University School of Medicine.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

The cells (80,000) were incubated with the Cy5.5-labeled polymer for 4 h, washed twice with PBS and treated with trypsin, and then collected for measurement. Details are provided in Methods and in the figure legends.

Instrument

The data were acquired using FACS Calibur (BD).

Software

Data acquired by FACS Calibur was analyzed using BD CellQuest Pro Software (version 5.1) and FlowJo_V10 software (version 10.7.1).

Cell population abundance

No sorting was performed.

Gating strategy

By adjusting the FSC/SSC voltage, the live-cell groups (gate1) were acquired for analysis, and the acquisition stopped when 10,000 cells among Gate 1 were counted. Our histogram plots a single parameter (Cy5.5 fluorescent intensity, horizontal axis) against the number of events detected (vertical axis). Negative control cells were plotted in black (defined as Cy5.5 intensity lower than 10), and experimental cells stained by Cy5.5-labeled polymers were plotted in color.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.